2025

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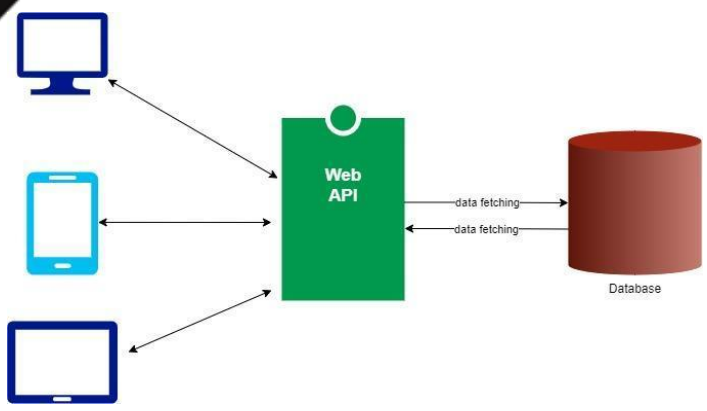
EXERCISE
Internet of Things
PROGRAMMING



MỤC LỤC

Setup Development Environment

Example 0.01

Mục tiêu: Thiết lập môi trường lập trình Internet of Things và tạo project

Yêu cầu:

- ✓ Cài đặt JDK và Visual Studio Code
- ✓ Tạo project có tên project là **example01**

Hướng dẫn:

Bước 1: Cài đặt JDK và Visual Studio Code

JDK Development Kit 17.0.7 downloads

JDK 17 binaries are free to use in production and free to redistribute, at no cost, under the Oracle No-Fee Terms and Conditions.

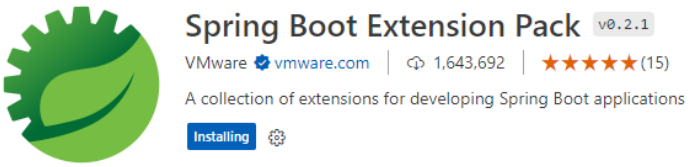
JDK 17 will receive updates under these terms, until September 2024, a year after the release of the next LTS.

Linux **macOS** **Windows**

Product/file description	File size	Download
x64 Compressed Archive	172.19 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.zip (sha256)
x64 Installer	153.28 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.exe (sha256)
x64 MSI Installer	152.07 MB	https://download.oracle.com/java/17/latest/jdk-17_windows-x64_bin.msi (sha256)

Bước 2: Để cài đặt Extension Java cho Visual Studio Code bạn hãy thực hiện Mở Extensions (Ctrl+Shift+X), tìm kiếm **java**

Bước 3: Để cài đặt Extension Spring Boot cho Visual Studio Code bạn hãy thực hiện Mở Extensions (Ctrl+Shift+X), tìm kiếm **Spring Boot Extension Pack**

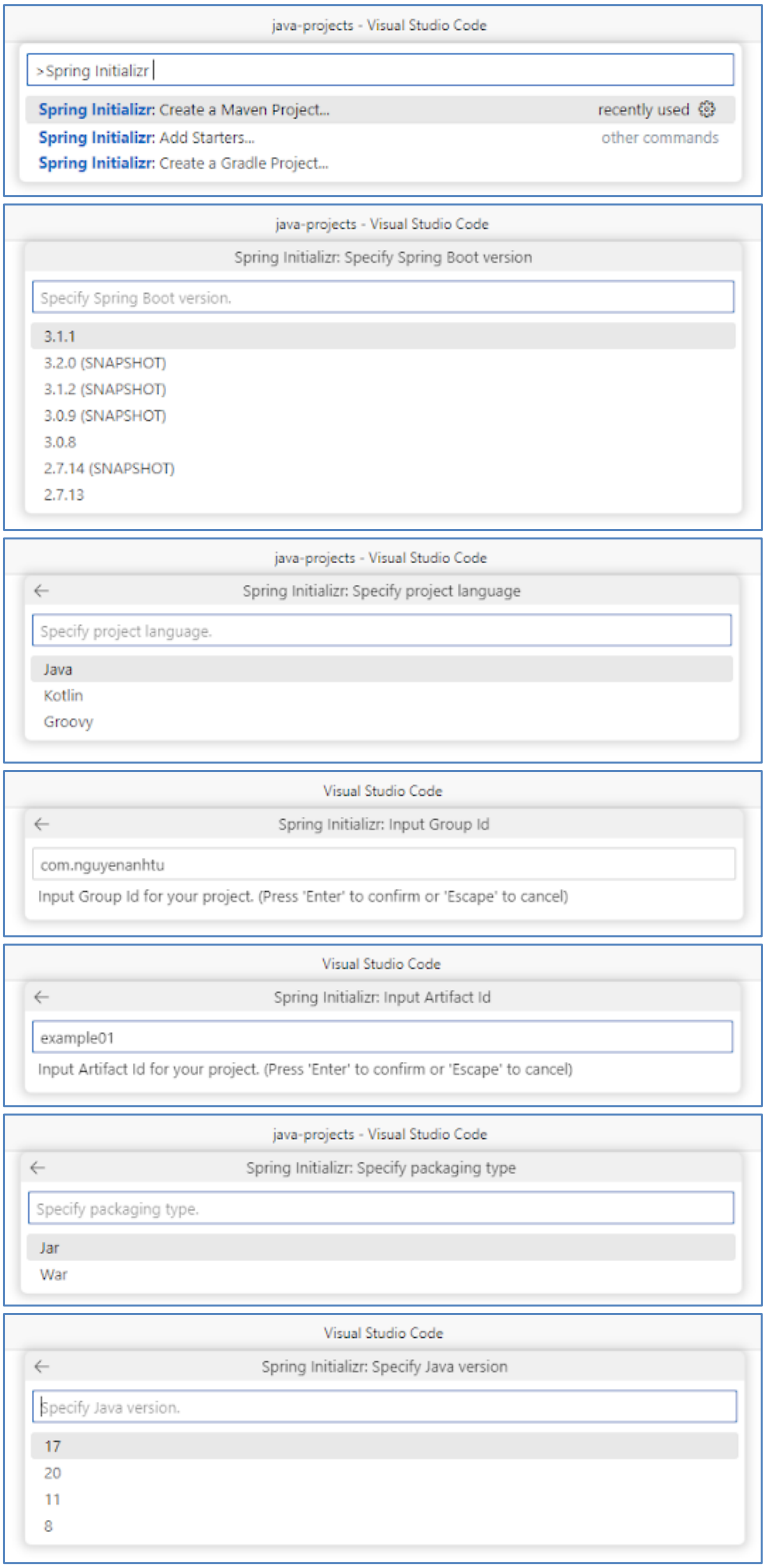


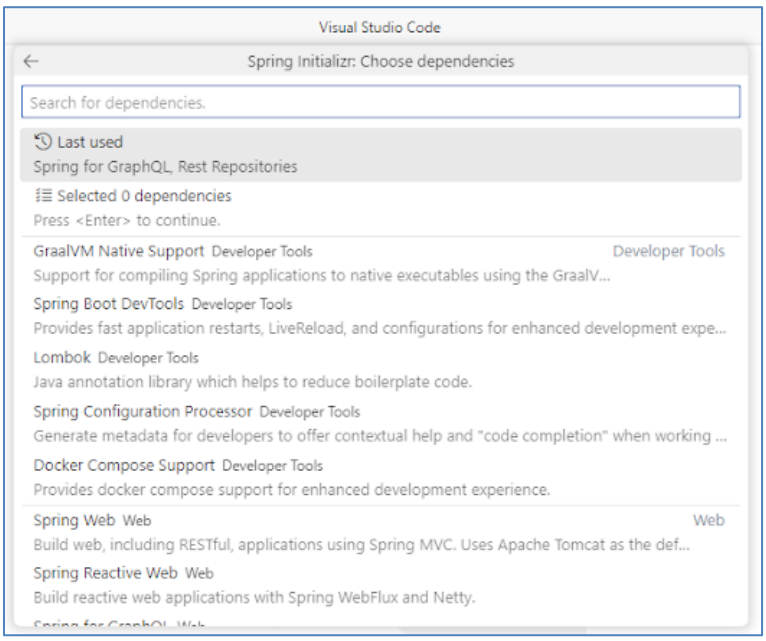
Bước 4: Sử dụng Visual Studio Code tạo project **Maven** với thông tin như sau:

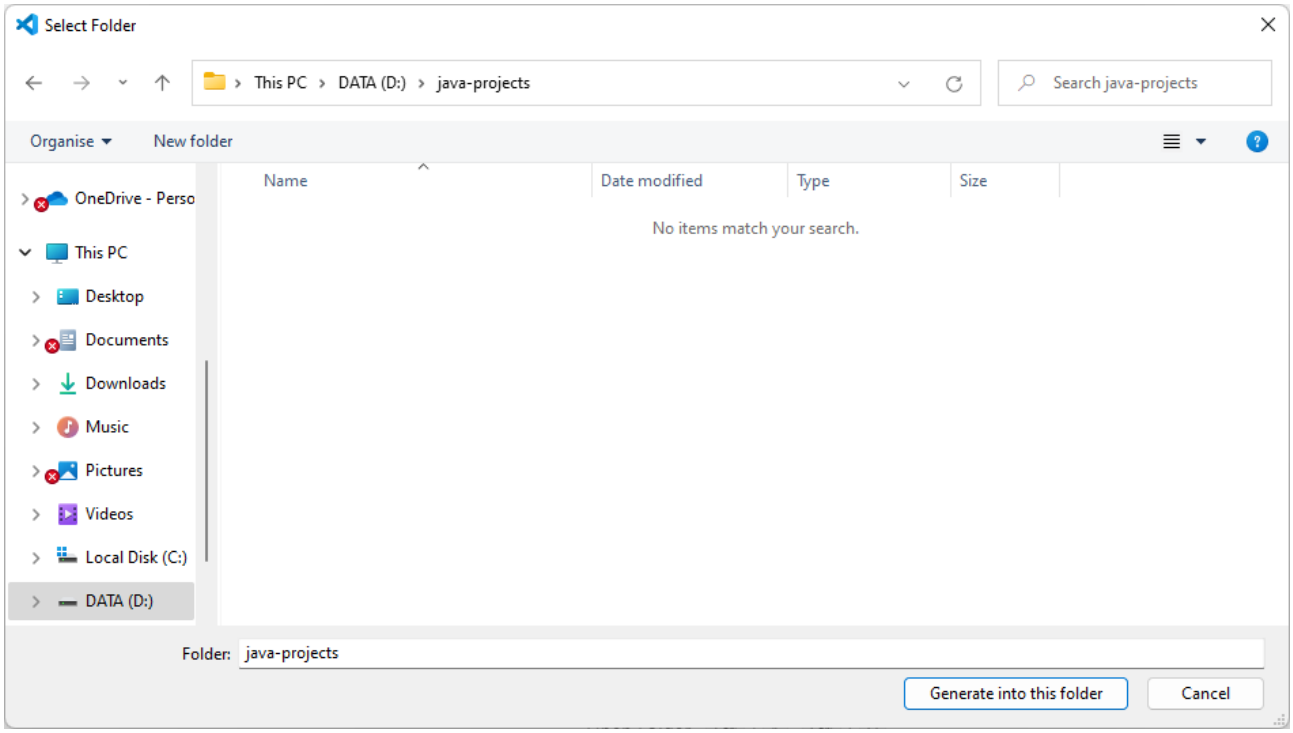
- ✓ Spring Boot version: **3.x.x**
- ✓ Language: **Java**
- ✓ Group Id: **com.nguyenanhtu**
- ✓ Artifact Id: **example01**
- ✓ Packaging type: **Jar**
- ✓ Java Version: **17**
- ✓ Dependencies: **Selected 0 dependencies**

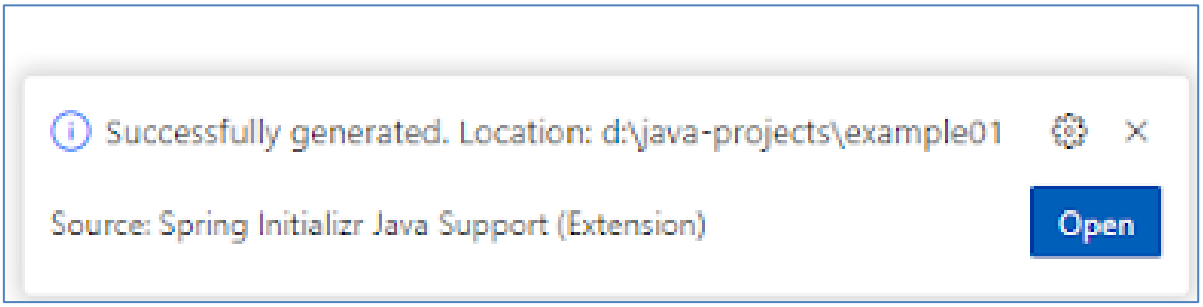
✓ Vị trí project: **D:\java-projects**

Hãy mở Bảng lệnh (Ctrl+Shift+P) và nhập **Spring Initializr** để bắt đầu tạo dự án **Maven**

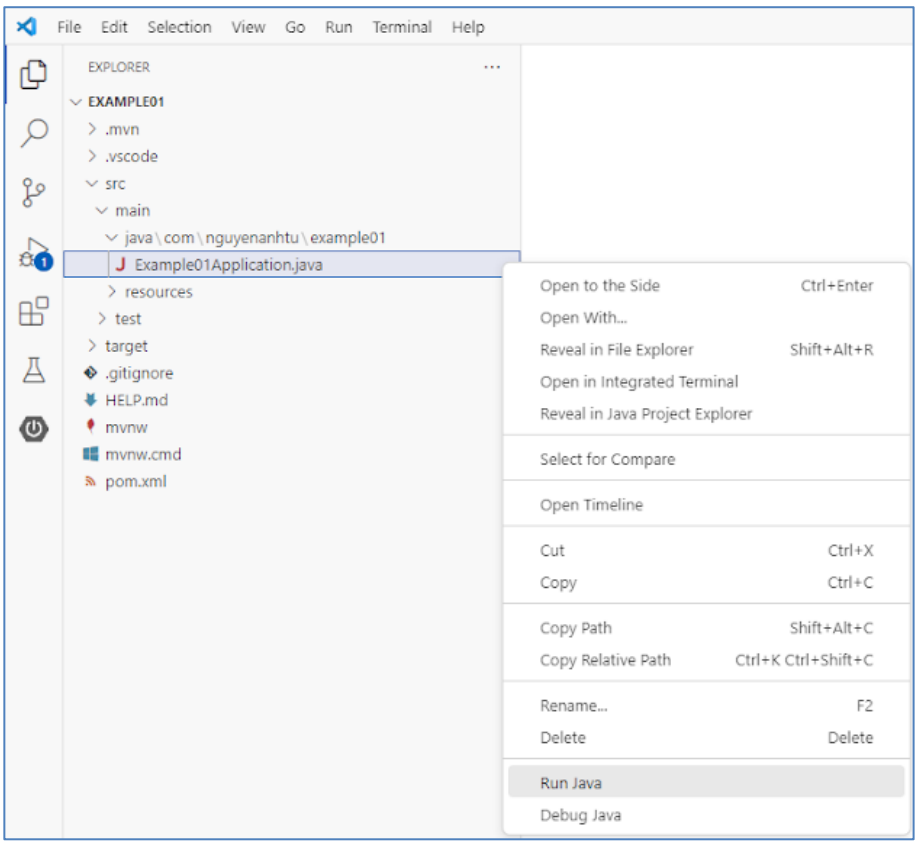


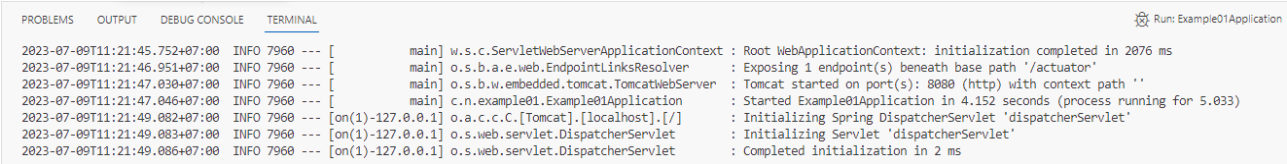






Bước 5: Click chuột phải vào lớp **Example01Application** sau đó Click vào **Run Java**.

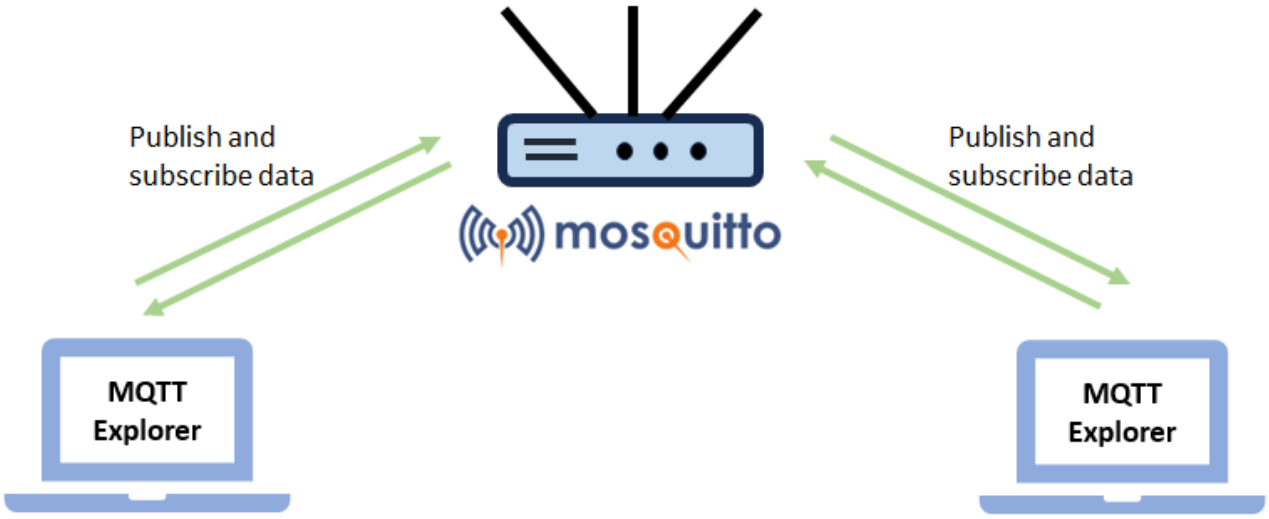

File Edit Selection View Go Run Terminal Help
EXPLORER
EXAMPLE01
.mvn
.vscode
src
main
java \ com \ nguyenanhtu \ example01
Example01Application.java
resources
test
target
.gitignore
HELP.md
mvnw
mvnw.cmd
pom.xml
Open to the Side Ctrl+Enter
Open With...
Reveal in File Explorer Shift+Alt+R
Open in Integrated Terminal
Reveal in Java Project Explorer
Select for Compare
Open Timeline
Cut Ctrl+X
Copy Ctrl+C
Copy Path Shift+Alt+C
Copy Relative Path Ctrl+K Ctrl+Shift+C
Rename... F2
Delete Delete
Run Java
Debug Java


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
Run: Example01Application
2023-07-09T11:21:45.752+07:00 INFO 7960 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2076 ms
2023-07-09T11:21:46.951+07:00 INFO 7960 --- [main] o.s.b.a.e.web.EndpointLinksResolver : Exposing 1 endpoint(s) beneath base path '/actuator'
2023-07-09T11:21:47.030+07:00 INFO 7960 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8080 (http) with context path ''
2023-07-09T11:21:47.046+07:00 INFO 7960 --- [main] c.n.example01.Example01Application : Started Example01Application in 4.152 seconds (process running for 5.033)
2023-07-09T11:21:49.082+07:00 INFO 7960 --- [on(1)-127.0.0.1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2023-07-09T11:21:49.083+07:00 INFO 7960 --- [on(1)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2023-07-09T11:21:49.086+07:00 INFO 7960 --- [on(1)-127.0.0.1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms

IoT Broker

Example 1.01

Mục tiêu: Quản lý ứng dụng **MQTT-Explorer** kết nối **MQTT Broker** có kiến trúc như sau:



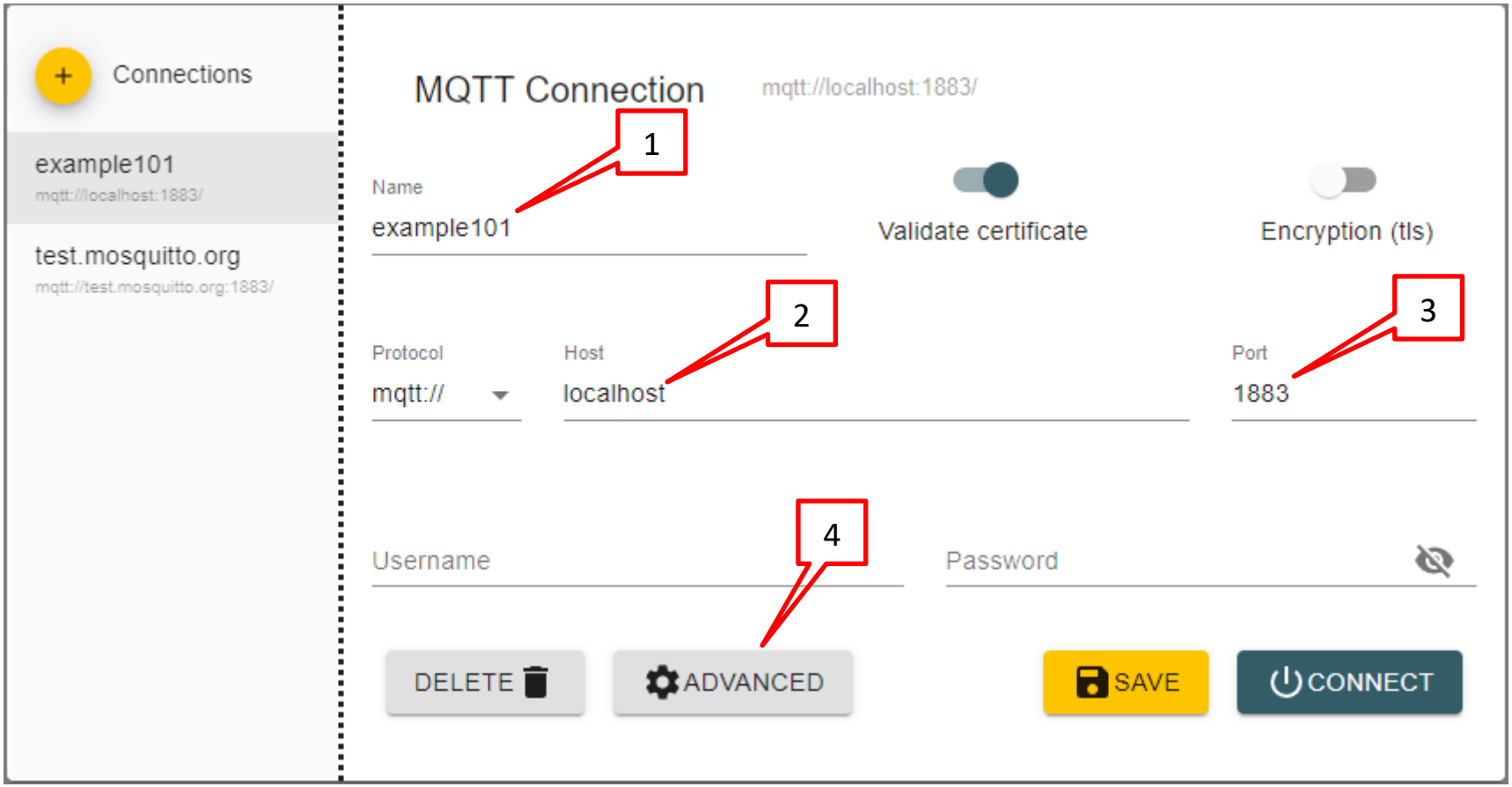
Yêu cầu: Thực hiện các chức năng sau:

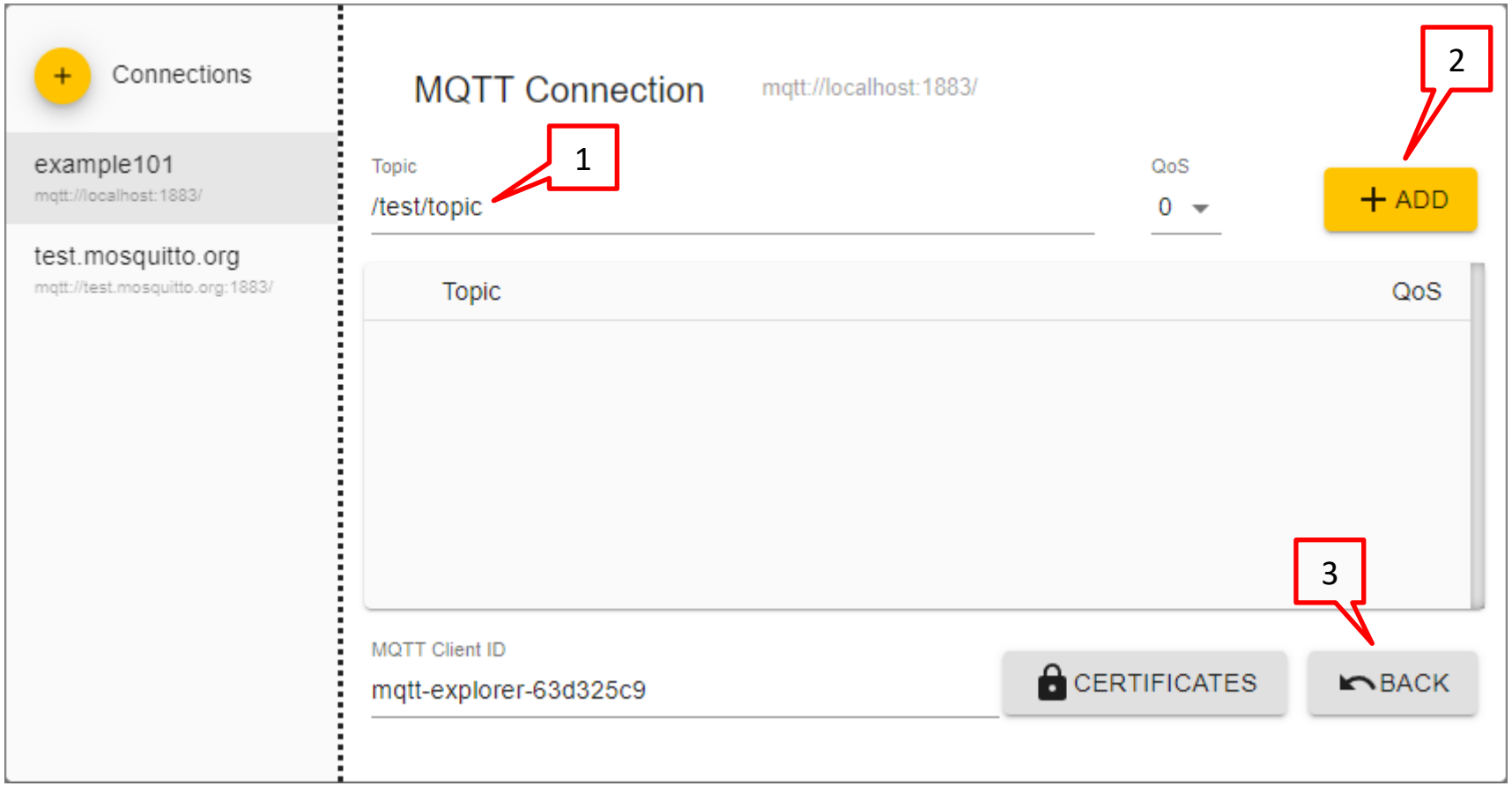
- ✓ Sử dụng **MQTT-Explorer** Subscribe với **Broker** với topic có tên */test/topic*
- ✓ Sử dụng **MQTT-Explorer** Publish nội dung "Hi from the IoT application" đến **Broker** với topic có tên */test/topic*

Hướng dẫn:

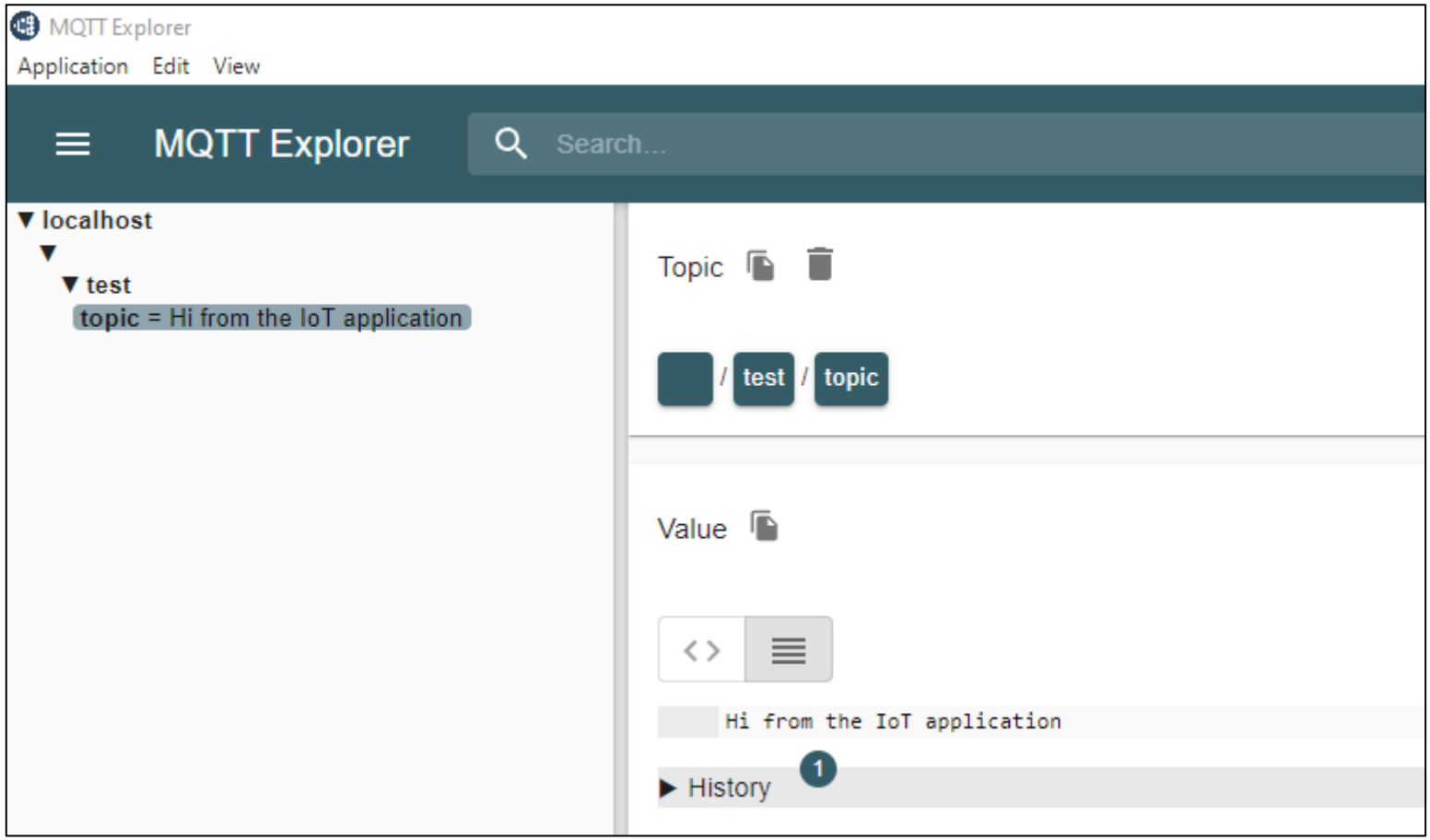
Bước 1: Sử dụng Visual Studio Code, mở Command Palette (Ctrl+Shift+P) và nhập **Java: Create Java Project…)** để bắt đầu tạo project với các thông tin như sau:

Bước 1: Sử dụng **MQTT-Explorer** đăng ký topic có tên **/test/topic** với Broker





Bước 4: Sử dụng **MQTT-Explorer** publish tới Broker nội dung **"Hi from the IoT application"** với topic có tên **/test/topic**


MQTT Explorer
Application Edit View
MQTT Explorer
Search...
localhost
test
topic = Hi from the IoT application
Topic
/ test / topic
Value
Hi from the IoT application
History
1

Example 1.02

Mục tiêu: Tạo và quản lý ứng dụng **Java** sử dụng **MQTT Paho** kết nối **MQTT Broker** có kiến trúc như sau:

Yêu cầu: Ứng dụng **Java** Thực hiện các chức năng sau:

- ✓ Subscribe với **Broker** với topic có tên */test/topic*
- ✓ Publish nội dung "Hi from the IoT application" đến **Broker** với topic có tên */test/topic*

Hướng dẫn:

Bước 2: Sửa file pom.xml như sau:

pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.nguyenanhtu</groupId>
    <artifactId>example102</artifactId>
    <version>1.0-SNAPSHOT</version>

    <dependencies>
      <dependency>
            <groupId>org.eclipse.paho</groupId>
            <artifactId>org.eclipse.paho.client.mqttv3</artifactId>
            <version>1.2.5</version>
      </dependency>
    </dependencies>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>
</project>
```

Bước 4: Tao Entity **Employee** như sau:

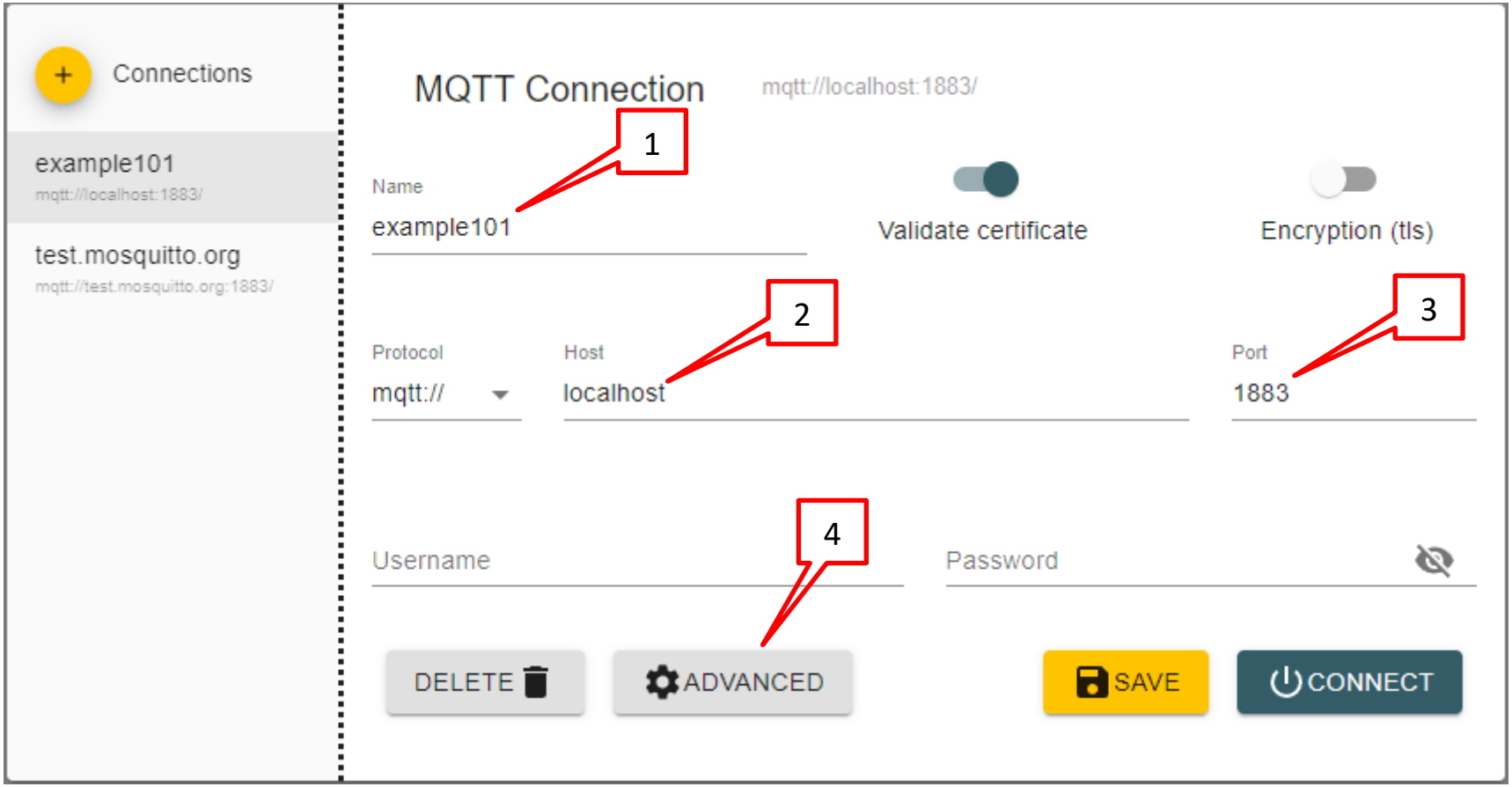
Main.java

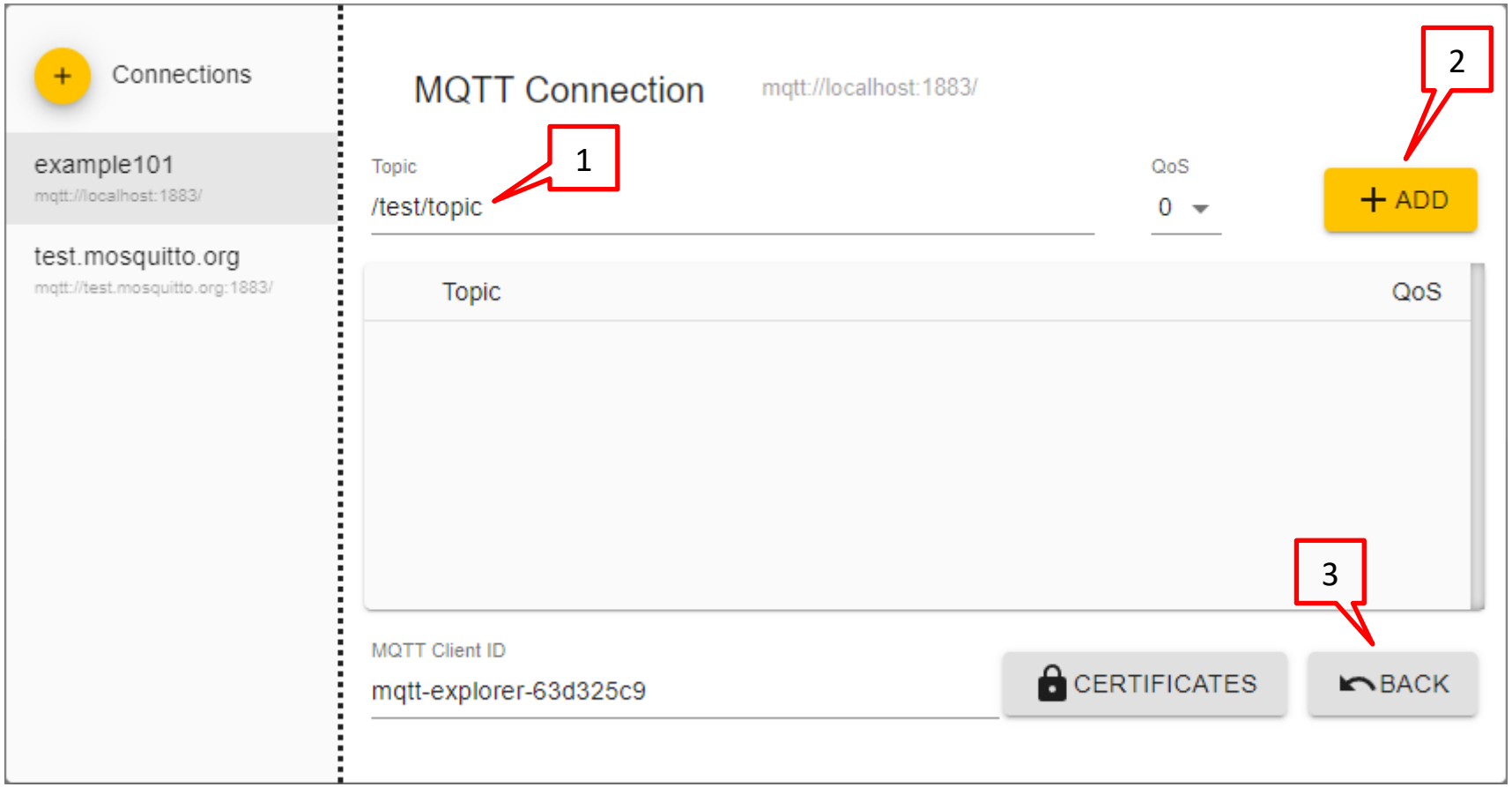
```
package com.nguyenanhtu.example101;

import java.io.IOException;

import org.eclipse.paho.client.mqttv3.IMqttDeliveryToken;
import org.eclipse.paho.client.mqttv3.MqttCallback;
import org.eclipse.paho.client.mqttv3.MqttException;
import org.eclipse.paho.client.mqttv3.MqttMessage;
import org.eclipse.paho.client.mqttv3.MqttConnectOptions;
import org.eclipse.paho.client.mqttv3.MqttClient;

public class Main {
    public static void main(String[] args) {
        String mqttBroker = "tcp://localhost:1883";
        String mqttTopic = "/test/topic";
        String username = "IoTClient";
        String password = "IoTPass";
        String testMsg = "Hi from the IoT application";
        int qos = 1;

        try {
            MqttClient mqttClient = new MqttClient(mqttBroker, "mqttClient");
            MqttConnectOptions mqttOptions = new MqttConnectOptions();
            mqttOptions.setUserName(username);
            mqttOptions.setPassword(password.toCharArray());
            mqttClient.connect(mqttOptions);

            if (mqttClient.isConnected()) {
                mqttClient.setCallback(new MqttCallback() {
                    @Override
```

```
                public void messageArrived(String topic, MqttMessage message) throws Exception {
                    System.out.println("Received message: " + new String(message.getPayload()));
                }

                @Override
                public void connectionLost(Throwable cause) {
                    System.out.println("Connection is lost: " + cause.getMessage());
                }

                @Override
                public void deliveryComplete(IMqttDeliveryToken token) {
                    System.out.println("Message publish is complete: " + token.isComplete());
                }
            });

            /* The client subscribe to a topic */
            mqttClient.subscribe(mqttTopic, qos);
            /* Preparing a message to be published */
            MqttMessage mqttMsg = new MqttMessage(testMsg.getBytes());
            mqttMsg.setQos(qos);
            /* A message is published on the same subscribed topic */
            mqttClient.publish(mqttTopic, mqttMsg);
        }

        /* Keep the application open, so that the subscribe operation can tested */
        System.out.println("Press Enter to disconnect");
        System.in.read();
        /* Proceed with disconnecting */
        mqttClient.disconnect();
        mqttClient.close();

    } catch (MqttException e) {
        e.printStackTrace();
```

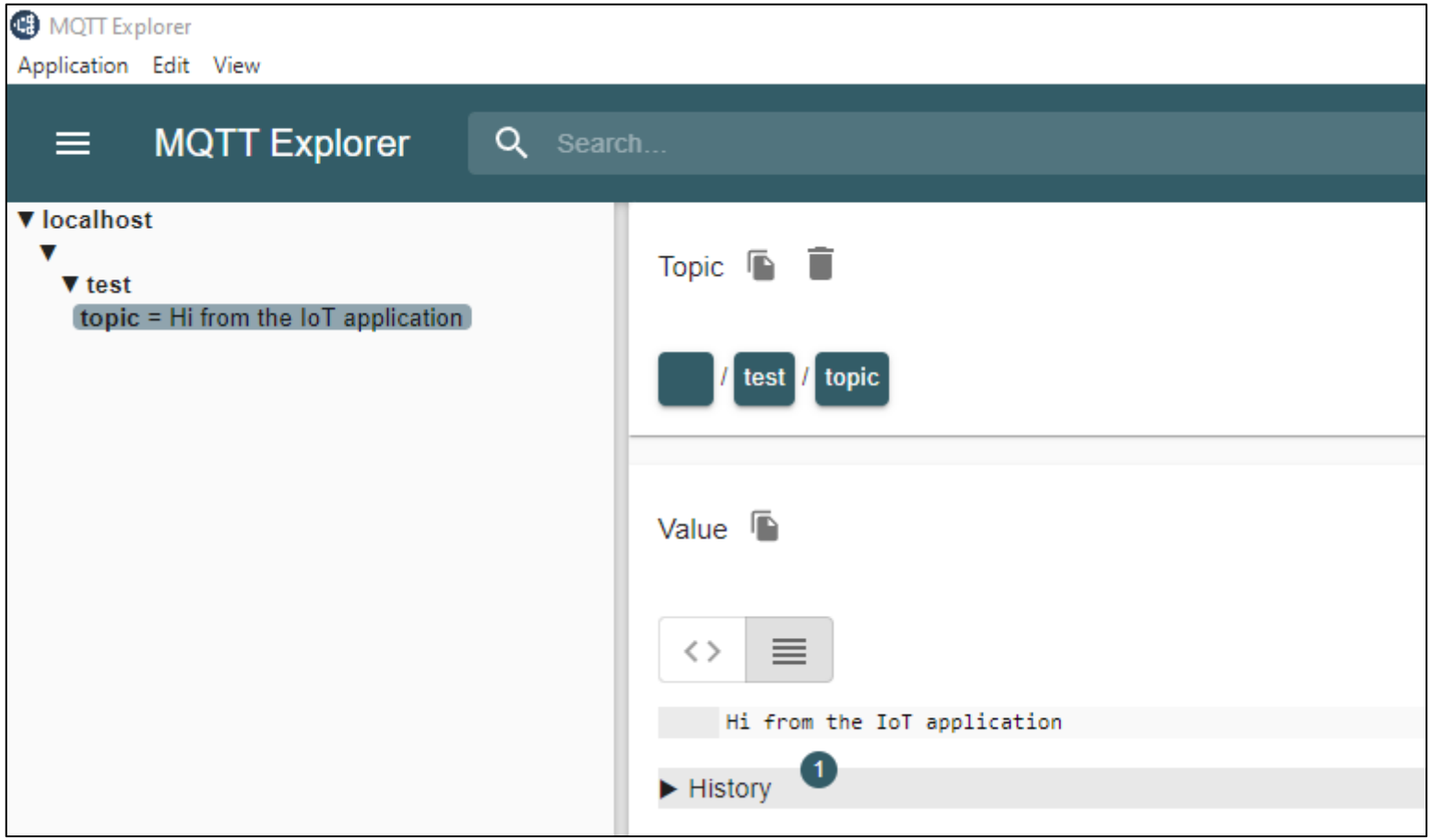
```
        } catch (IOException e) {
            throw new RuntimeException(e);
        }
    }
}
```

Bước 3: Sử dụng **MQTT-Explorer** đăng ký topic có tên **/test/topic** với Broker

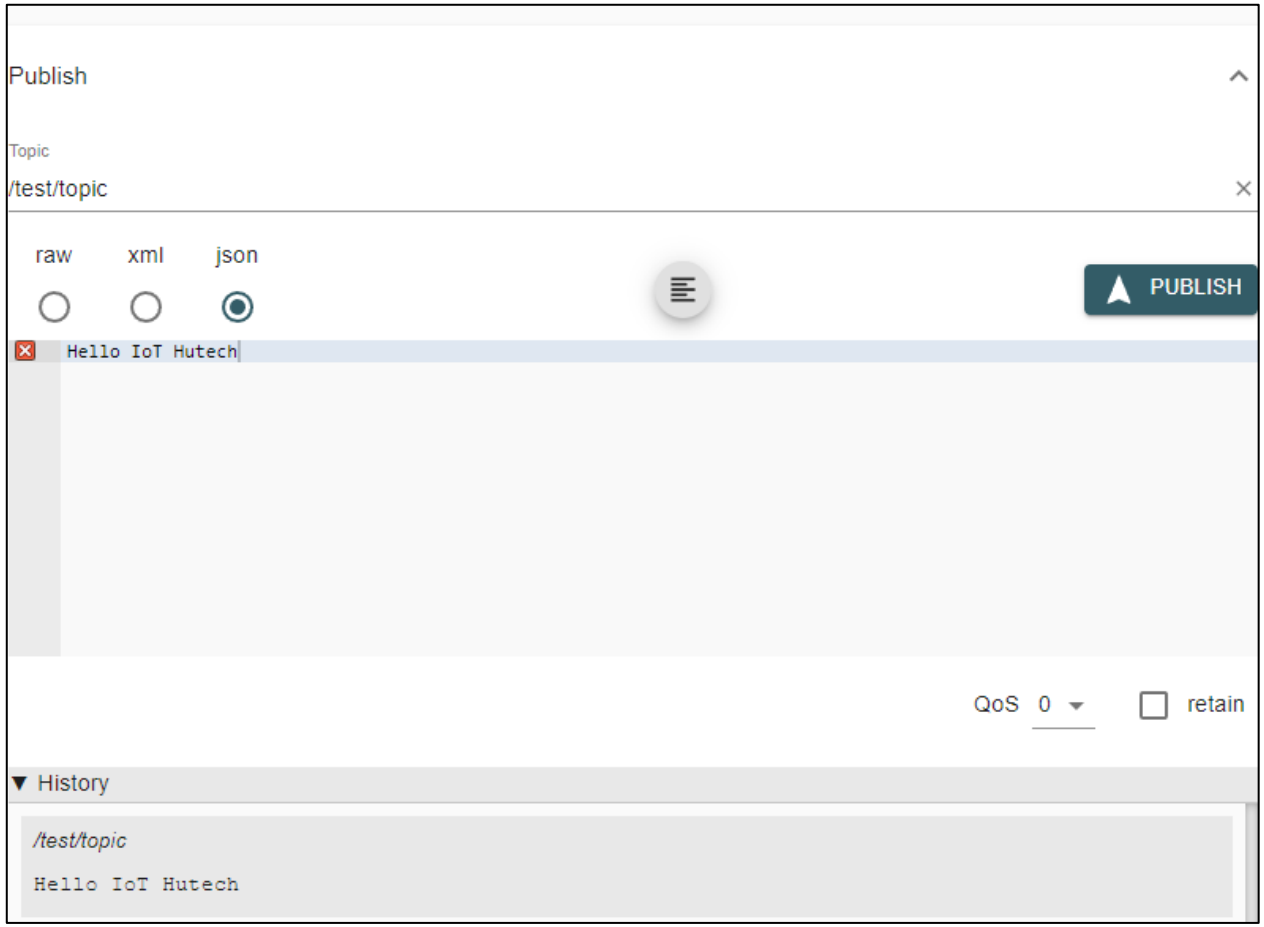




Bước 4: Click chuột phải vào lớp **Main →Run Java,** sau đó quan sát ứng dụng **MQTT-Explorer** như sau**:**



Bước 4: Sử dụng **MQTT-Explorer** publish tới Broker nội dung **Hello IoT Hutech** với topic có tên **/test/topic**

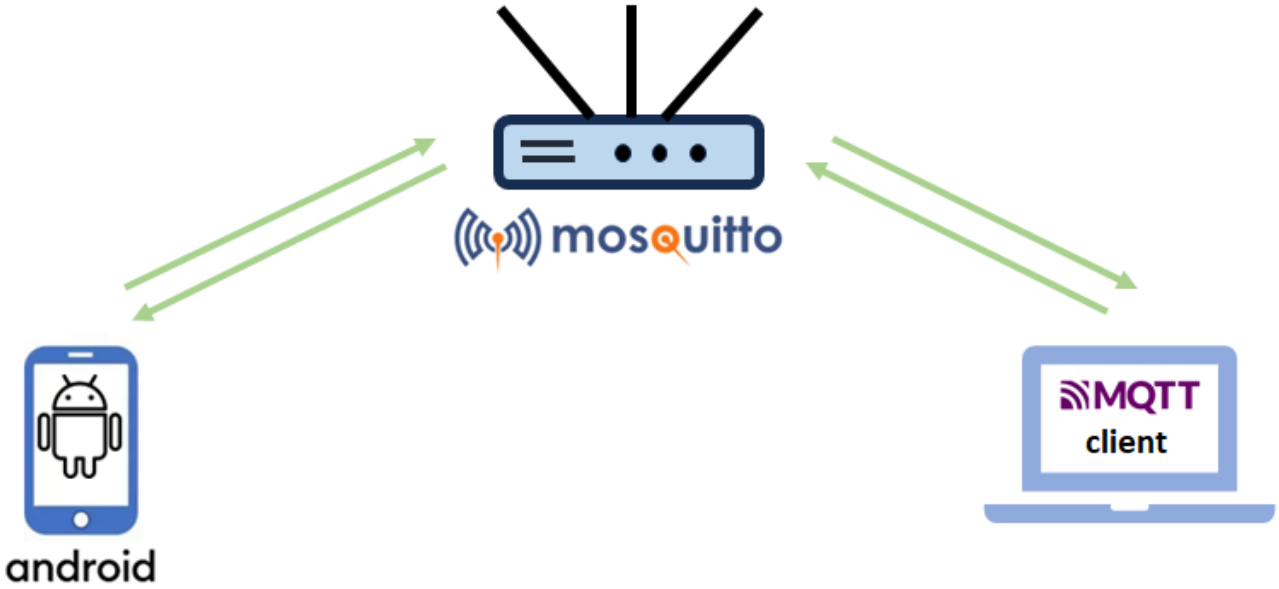


Bước 5: Kiểm tra dữ liệu nhận được trong cửa sổ Terminal của Visual Code

```
Press Enter to disconnect
Received message: Hi from the IoT application
Message publish is complete: true
Received message: Hello IoT Hutech
```

Example 1.03

Mục tiêu: Tạo và quản lý ứng dụng **Android** sử dụng **MQTT (Eclipse Paho Java MQTT client)** kết nối đến **MQTT Broker** có topic tên **/sensor/temp**



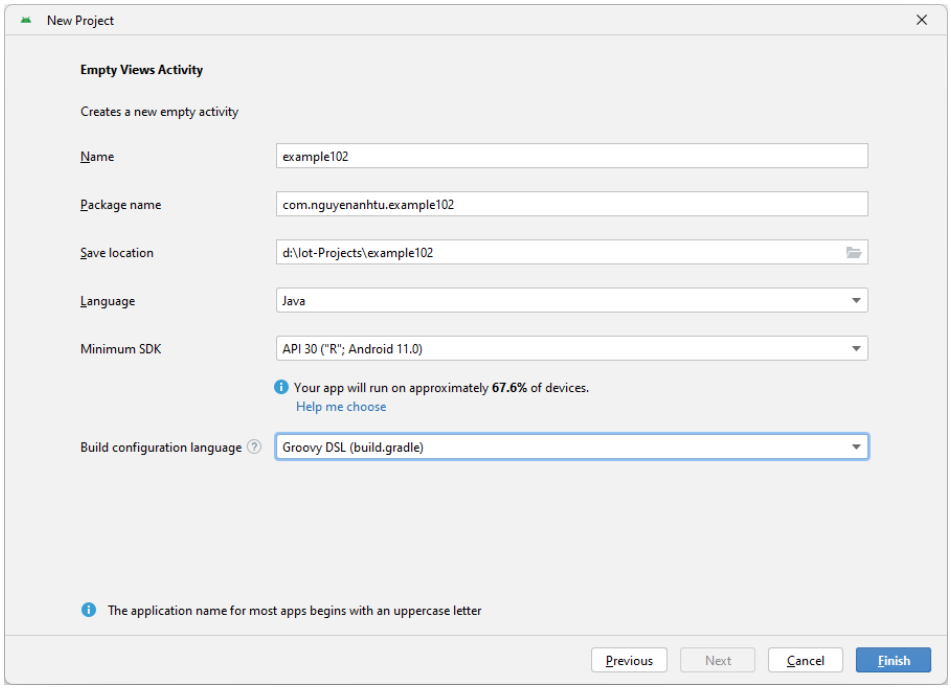
Yêu cầu: Ứng dụng **Android** Thực hiện các chức năng sau:

- ✓ Subscribe với **Broker** với topic có tên **/sensor/temp**

Hướng dẫn:

Bước 1: Sử dụng Android Studio tạo project với các thông tin như sau:

- ✓ Build configuration language: **Groovy DSL(build.gradle)**
- ✓ Vị trí project: **D:\IoT-projects**
- ✓ Artifact Id: **example102**
- ✓ Group Id: **com.nguyenanhtu**
- ✓ Language: **Java**



Bước 2: Thêm vào file **build.gradle** như sau:

build.gradle

```
dependencies {
      . . .
      implementation 'org.eclipse.paho:org.eclipse.paho.client.mqttv3:1.2.5'
      implementation 'org.eclipse.paho:org.eclipse.paho.android.service:1.1.1'
}
```

Bước 3: Thêm vào file **AndroidManifest.xml** như sau:

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.INTERNET" />
    <uses-permission android:name="android.permission.WAKE_LOCK" />
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
    <uses-permission android:name="android.permission.ACCESS_WIFI_STATE" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Example102"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
```

```
        </activity>

            <service android:name="org.eclipse.paho.android.service.MqttService" />

    </application>
</manifest>
```

Bước 4: Tạo file **MqttHelper.java** với nội dung như sau:

MqttHelper.java

```
package com.nguyenanhtu.example102;

import android.content.Context;
import android.util.Log;

import org.eclipse.paho.android.service.MqttAndroidClient;
import org.eclipse.paho.client.mqttv3.DisconnectedBufferOptions;
import org.eclipse.paho.client.mqttv3.IMqttActionListener;
import org.eclipse.paho.client.mqttv3.IMqttDeliveryToken;
import org.eclipse.paho.client.mqttv3.IMqttToken;
import org.eclipse.paho.client.mqttv3.MqttCallbackExtended;
import org.eclipse.paho.client.mqttv3.MqttConnectOptions;
import org.eclipse.paho.client.mqttv3.MqttException;
import org.eclipse.paho.client.mqttv3.MqttMessage;

public class MqttHelper {
    public MqttAndroidClient mqttAndroidClient;

    final String serverUri = "tcp://localhost:1883";

    final String clientId = "AndroidClient";
    final String subscriptionTopic = "/sensor/temp";

    final String username = "IoTClient";
    final String password = "IoTClient";

    public MqttHelper(Context context){
```

```
        mqttAndroidClient = new MqttAndroidClient(context, serverUri, clientId);
        mqttAndroidClient.setCallback(new MqttCallbackExtended() {
            @Override
            public void connectComplete(boolean b, String s) {
                Log.w("mqtt", s);
            }

            @Override
            public void connectionLost(Throwable throwable) {

            }

            @Override
            public void messageArrived(String topic, MqttMessage mqttMessage) throws Exception {
                Log.w("Mqtt", mqttMessage.toString());
            }

            @Override
            public void deliveryComplete(IMqttDeliveryToken iMqttDeliveryToken) {

            }
        });
        connect();
    }

    public void setCallback(MqttCallbackExtended callback) {
        mqttAndroidClient.setCallback(callback);
    }

    private void connect(){
        MqttConnectOptions mqttConnectOptions = new MqttConnectOptions();
        mqttConnectOptions.setAutomaticReconnect(true);
        mqttConnectOptions.setCleanSession(false);
        mqttConnectOptions.setUserName(username);
        mqttConnectOptions.setPassword(password.toCharArray());
```

```
        try {

            mqttAndroidClient.connect(mqttConnectOptions, null, new IMqttActionListener() {
                @Override
                public void onSuccess(IMqttToken asyncActionToken) {

                    DisconnectedBufferOptions disconnectedBufferOptions = new DisconnectedBufferOptions();
                    disconnectedBufferOptions.setBufferEnabled(true);
                    disconnectedBufferOptions.setBufferSize(100);
                    disconnectedBufferOptions.setPersistBuffer(false);
                    disconnectedBufferOptions.setDeleteOldestMessages(false);
                    mqttAndroidClient.setBufferOpts(disconnectedBufferOptions);
                    subscribeToTopic();
                }

                @Override
                public void onFailure(IMqttToken asyncActionToken, Throwable exception) {
                    Log.w("Mqtt", "Failed to connect to: " + serverUri + exception.toString());
                }
            });


        } catch (MqttException ex){
            ex.printStackTrace();
        }
    }


    private void subscribeToTopic() {
        try {
            mqttAndroidClient.subscribe(subscriptionTopic, 0, null, new IMqttActionListener() {
                @Override
                public void onSuccess(IMqttToken asyncActionToken) {
                    Log.w("Mqtt","Subscribed!");
                }
```

```
                @Override
                public void onFailure(IMqttToken asyncActionToken, Throwable exception) {
                    Log.w("Mqtt", "Subscribed fail!");
                }
            });

        } catch (MqttException ex) {
            System.err.println("Exception whilst subscribing");
            ex.printStackTrace();
        }
    }
}
```

Bước 5: Sửa file activity_main.xml như sau:

res\layout\activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView

        android:id="@+id/dataReceived"

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

Bước 6: Sửa file **MainActivity.java** như sau:

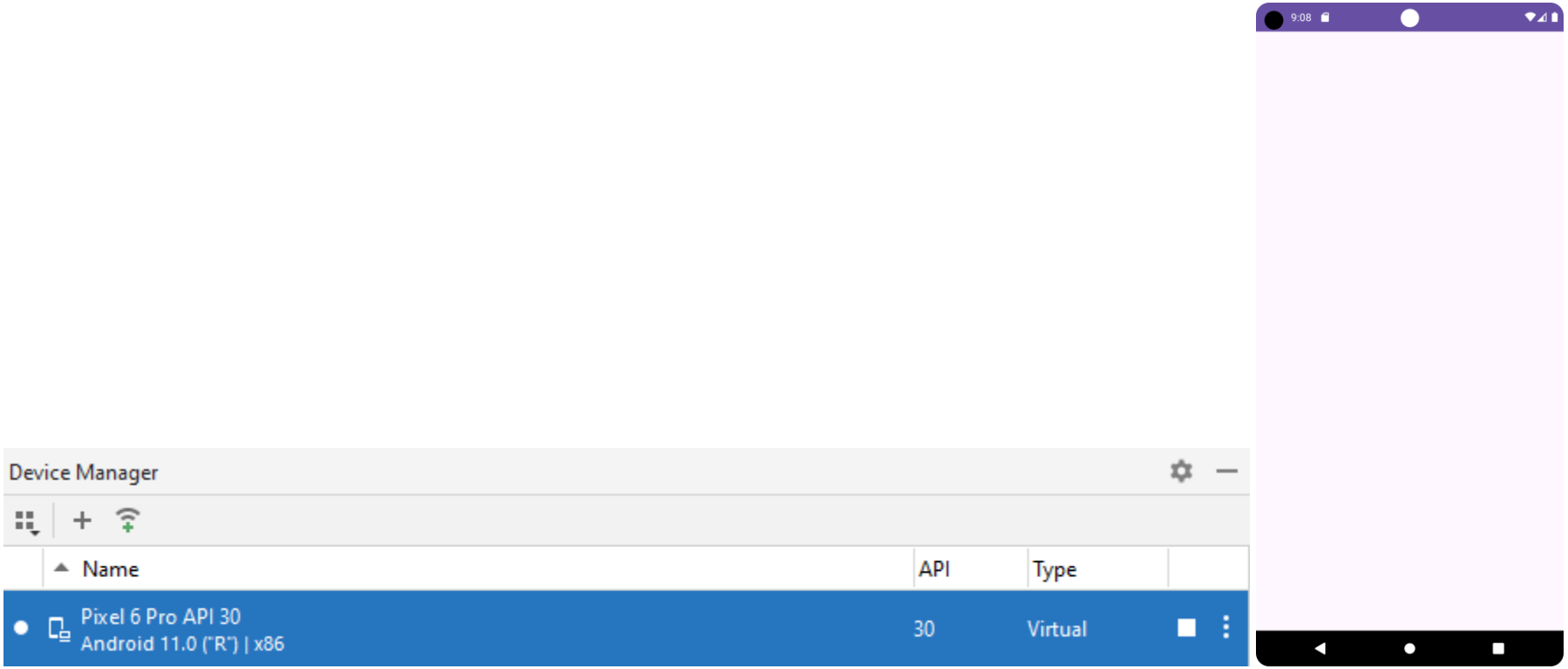
MainActivity.java

```
package com.nguyenanhtu.example102;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.util.Log;
import android.widget.TextView;

import org.eclipse.paho.client.mqttv3.IMqttDeliveryToken;
import org.eclipse.paho.client.mqttv3.MqttCallbackExtended;
import org.eclipse.paho.client.mqttv3.MqttMessage;

public class MainActivity extends AppCompatActivity {
    MqttHelper mqttHelper;
    TextView dataReceived;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        dataReceived = (TextView) findViewById(R.id.dataReceived);

        startMqtt();
    }

    private void startMqtt() {
        mqttHelper = new MqttHelper(getApplicationContext());
        mqttHelper.setCallback(new MqttCallbackExtended() {
            @Override
            public void connectComplete(boolean b, String s) {

            }

            @Override
            public void connectionLost(Throwable throwable) {

            }

            @Override
            public void messageArrived(String topic, MqttMessage mqttMessage) throws Exception {
```

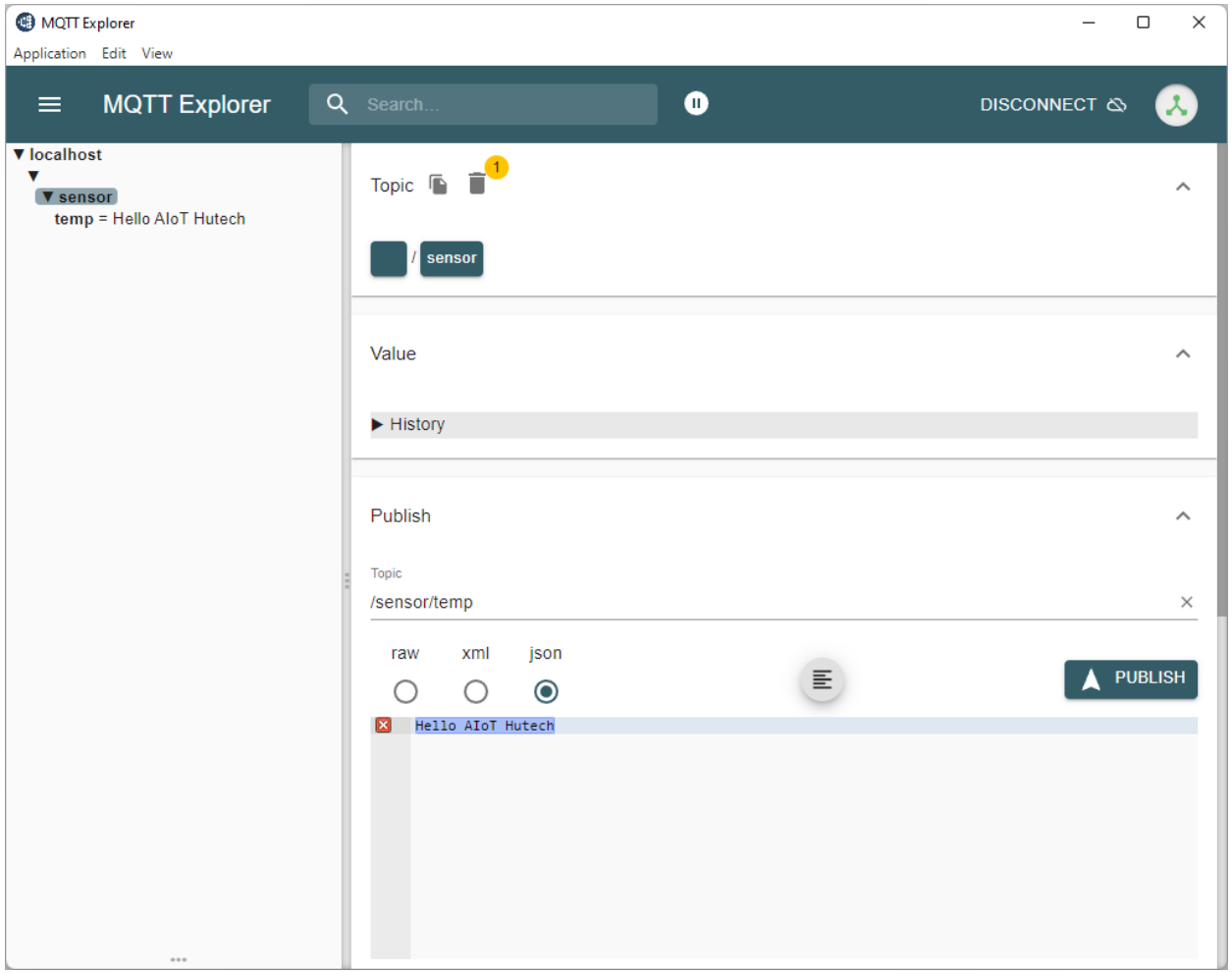
```
                Log.w("Debug", mqttMessage.toString());
                dataReceived.setText(mqttMessage.toString());
            }

            @Override
            public void deliveryComplete(IMqttDeliveryToken iMqttDeliveryToken) {

            }
        });
    }
}
```

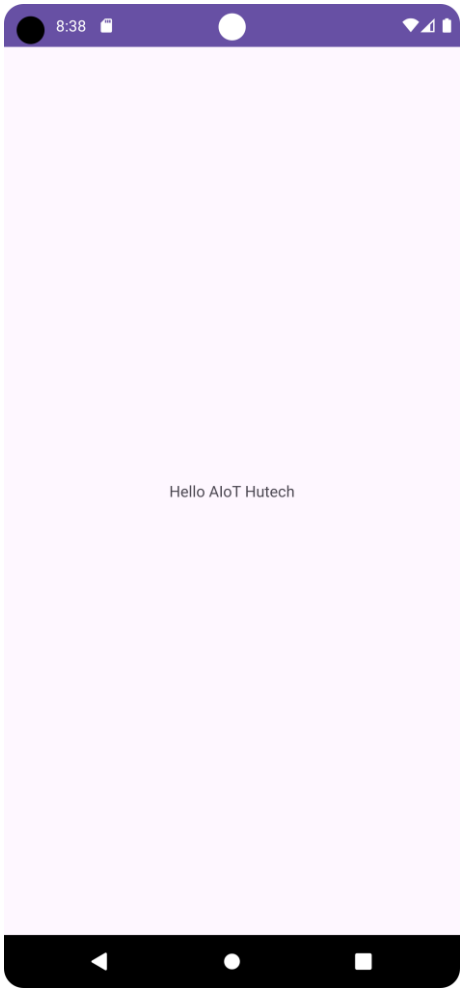
Bước 7: Sử dụng **MQTT-Explorer** subscribe với Broker một topic có tên **/sensor/temp**

Bước 8: Run App Android



Bước 9: Sử dụng **MQTT-Explorer** publish tới Broker nội dung **Hello AIoT Hutech** với topic có tên **/sensor/temp**





IoT Device

Example 2.01

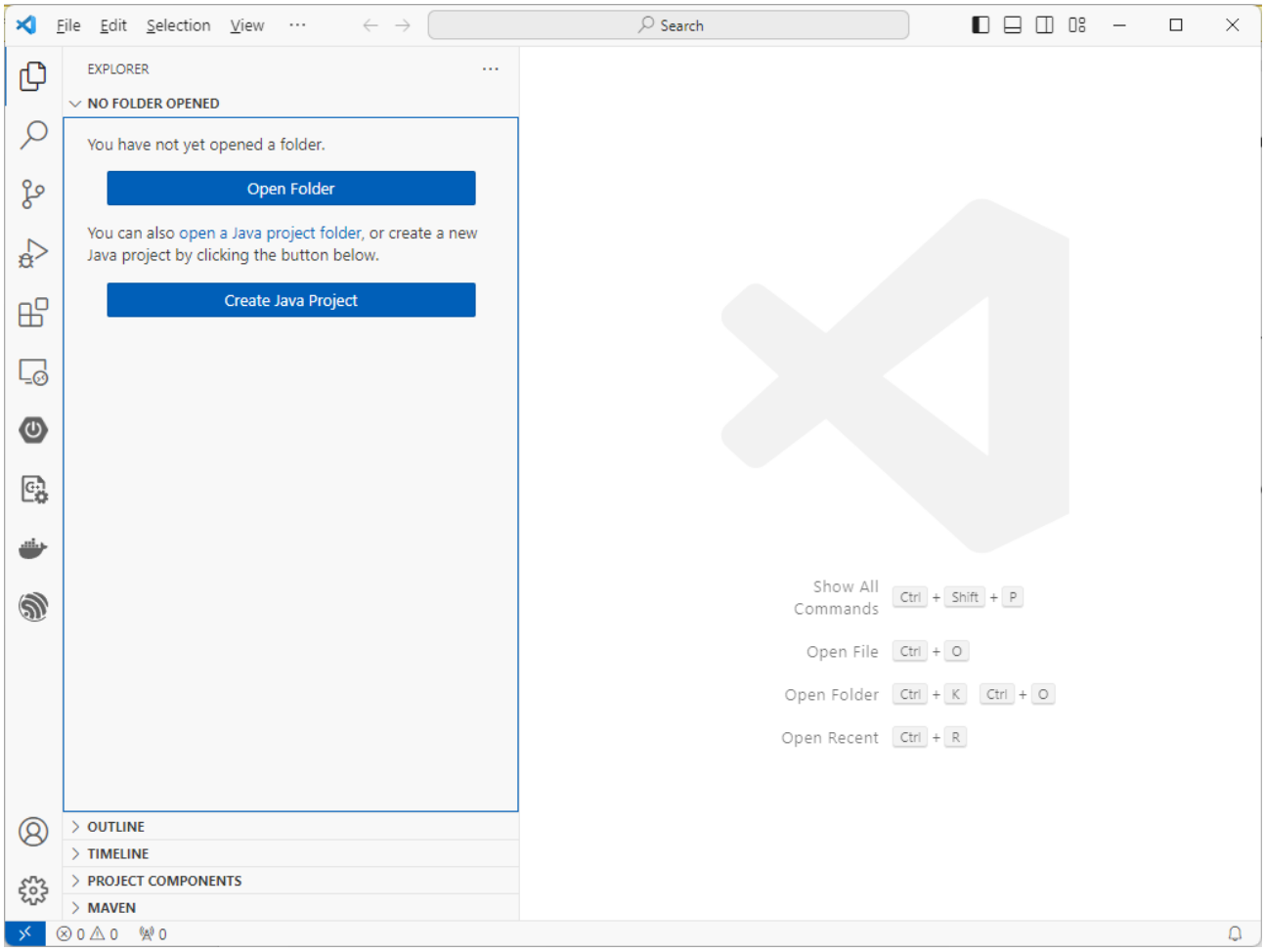
Mục tiêu: Thiết lập môi trường lập trình ESP-IDF (Espressif IoT Development Framework)

Yêu cầu:

- ✓ Cài đặt Visual Studio Code
- ✓ Cài đặt và cấu hình Espressif IDF Extentions

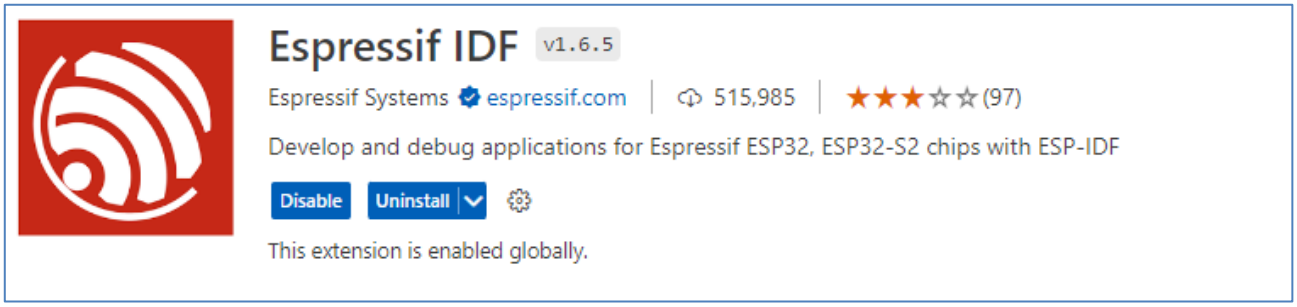
Hướng dẫn:

Bước 1: Download tại https://code.visualstudio.com, sau đó tiến hành cài đặt Visual Studio Code

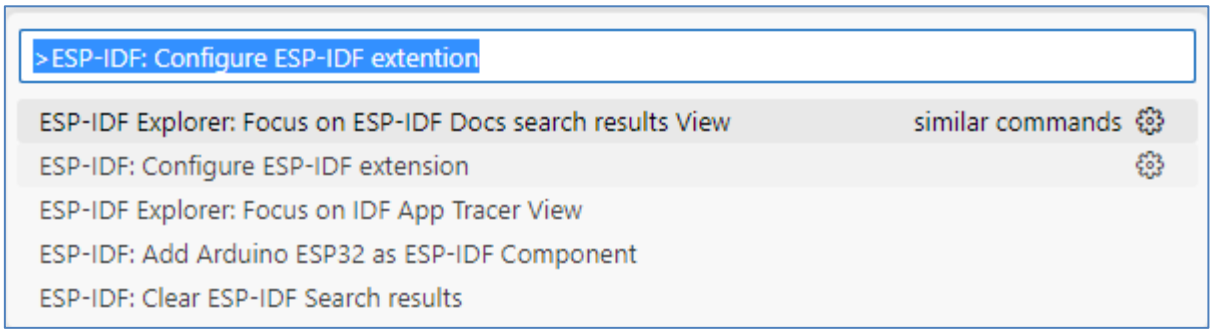


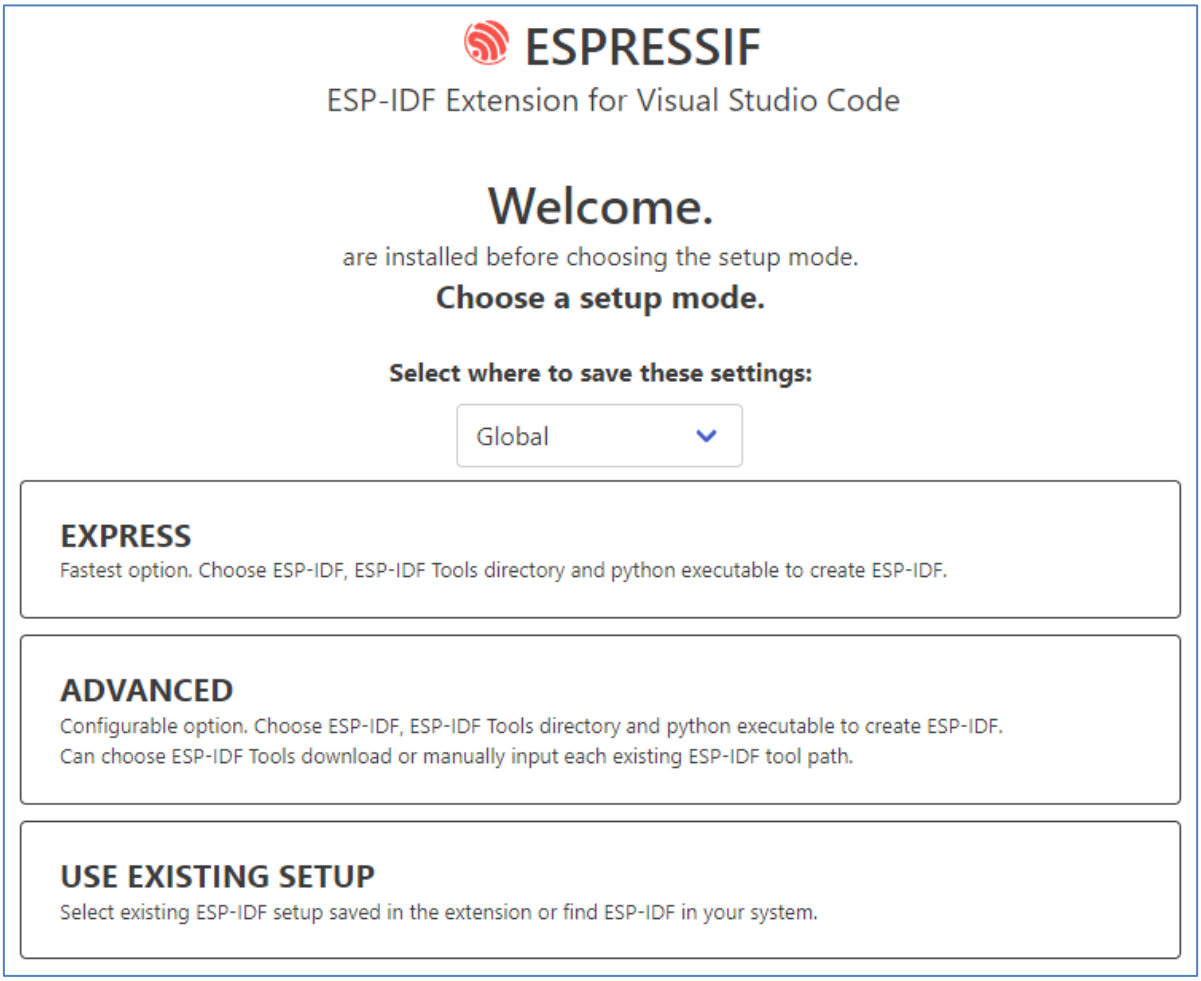
Bước 2: Cài đặt và cấu hình Espressif IDF

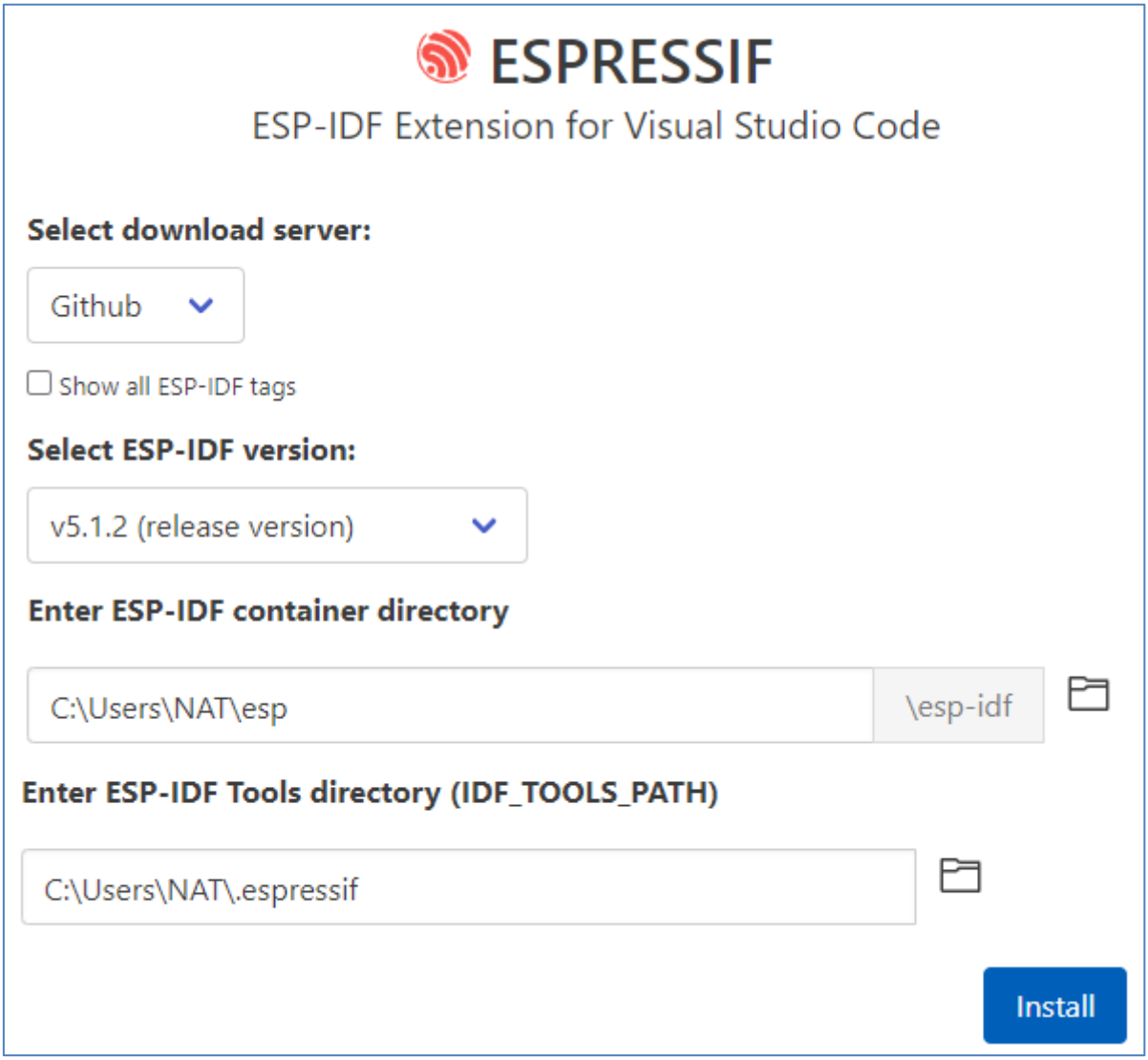
- ✓ Để cài đặt Extension Espressif IDF cho Visual Studio Code bạn hãy thực hiện Mở Extensions (Ctrl+Shift+X), tìm kiếm **Espressif IDF**



✓ Để cấu hình Extension Espressif IDF cho Visual Studio Code bạn hãy thực hiện Mở Command Palette (Ctrl+Shift+P), chọn **ESP-IDF: Configure ESP-IDF extention**






ESPRESSIF
ESP-IDF Extension for Visual Studio Code
Select download server:
Github
Show all ESP-IDF tags
Select ESP-IDF version:
v5.1.2 (release version)
Enter ESP-IDF container directory
C:\Users\NAT\esp
\esp-idf
Enter ESP-IDF Tools directory (IDF_TOOLS_PATH)
C:\Users\NAT\.espressif
Install

Example 2.02

Mục tiêu: Tạo và quản lý ứng dụng chạy trên vi điều khiển **ESP32**:

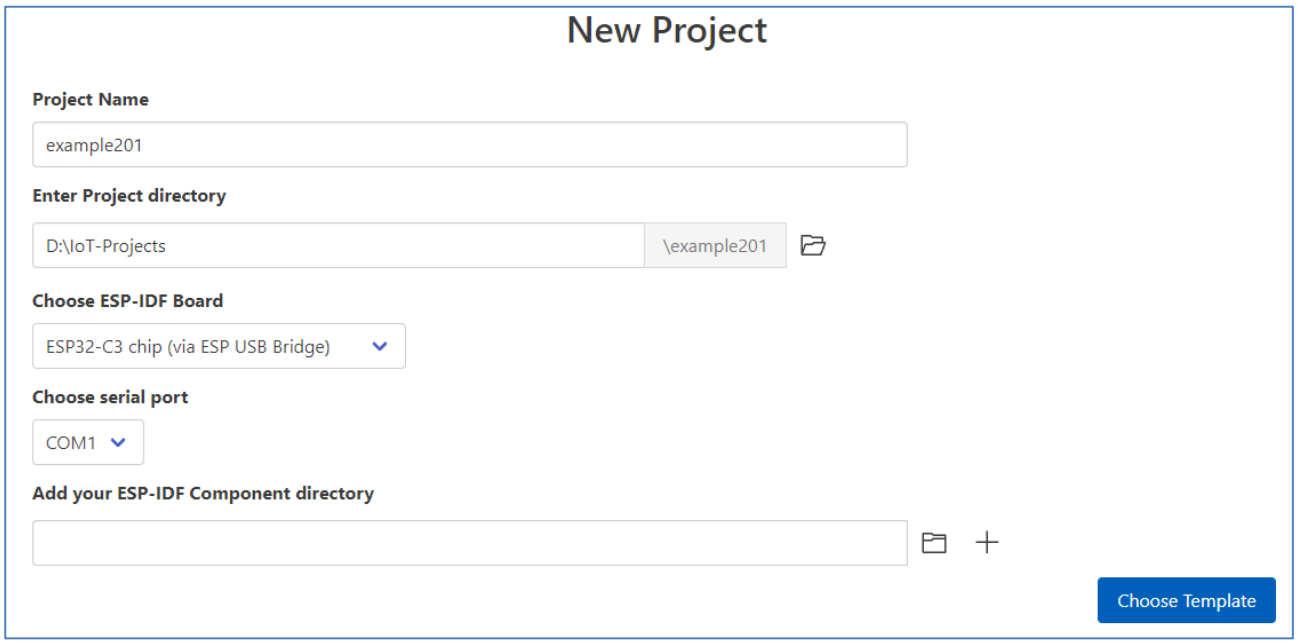
Yêu cầu: Ứng dụng thực hiện các chức năng sau:

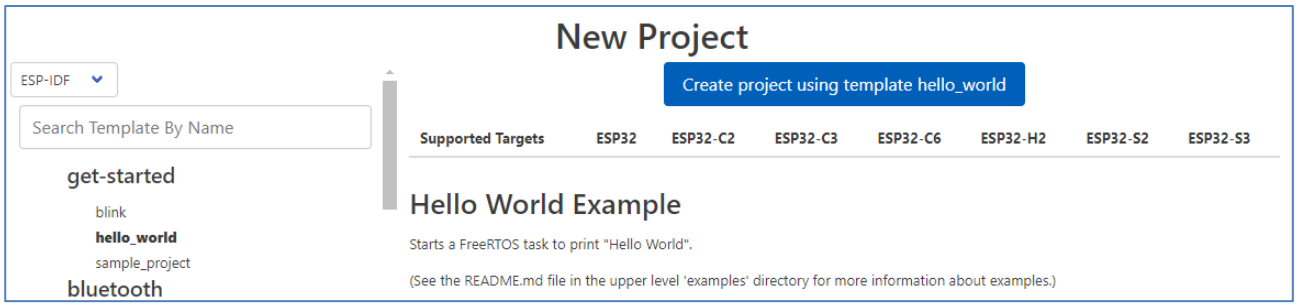
- ✓ In ra Teminal nội dung **Hello World**

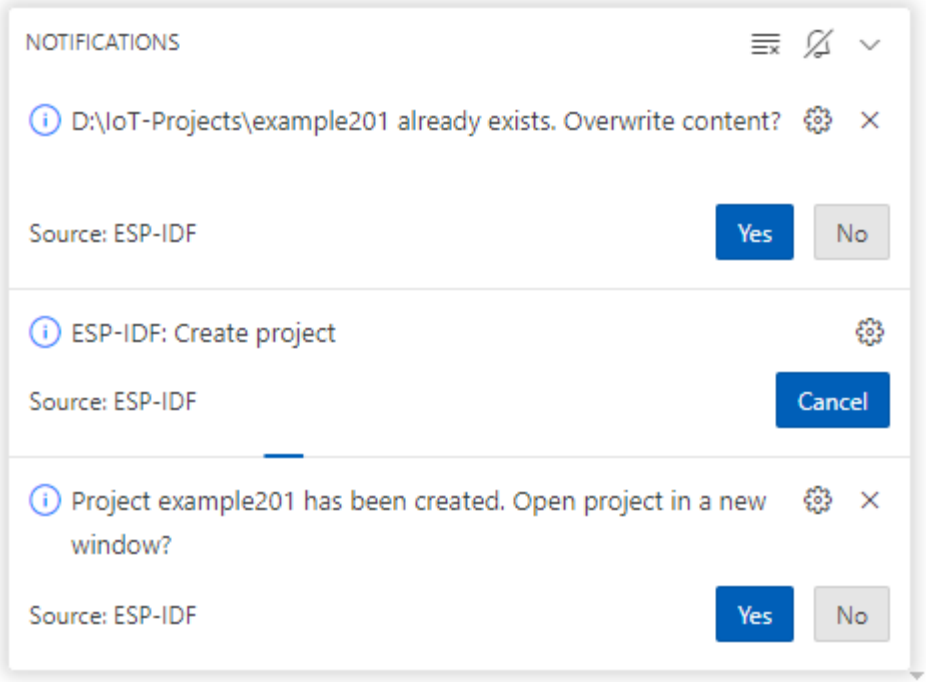
Hướng dẫn:

Bước 1: Sử dụng Visual Studio Code, mở Command Palette (Ctrl+Shift+P) và nhập **IDF: New Project)** để bắt đầu tạo project với các thông tin như sau:

- ✓ Project Name: **example202**
- ✓ Enter Project directory: **D:\java-projects**
- ✓ Choose ESP-IDF Board: **ESP32-C3 chip(via ESP USB Bridge)**
- ✓ Choose serial port: COMx
- ✓ Choose Template: **template-app**







Bước 2: Sửa file **hello_world_main.c** như sau:

hello_world_main.c

```
#include <stdio.h>
#include <inttypes.h>
#include "sdkconfig.h"
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "esp_chip_info.h"
#include "esp_flash.h"


void app_main(void)
```

```
{
    printf("Hello world!\n");
    printf("Restarting now.\n");
    fflush(stdout);
    esp_restart();
}
```

Bước 3: Build project và nạp chương trình vào vi điều khiển

Example 2.03

Mục tiêu: Tạo và quản lý ứng dụng sử dụng FreeRTOS chạy trên vi điều khiển **ESP32**:

Yêu cầu: Ứng dụng thực hiện các chức năng sau:

- ✓ In ra Teminal nội dung Hello World

Hướng dẫn:

Bước 1: Sử dụng Visual Studio Code, mở Command Palette (Ctrl+Shift+P) và nhập **IDF: New Project)** để bắt đầu tạo project với các thông tin như sau:

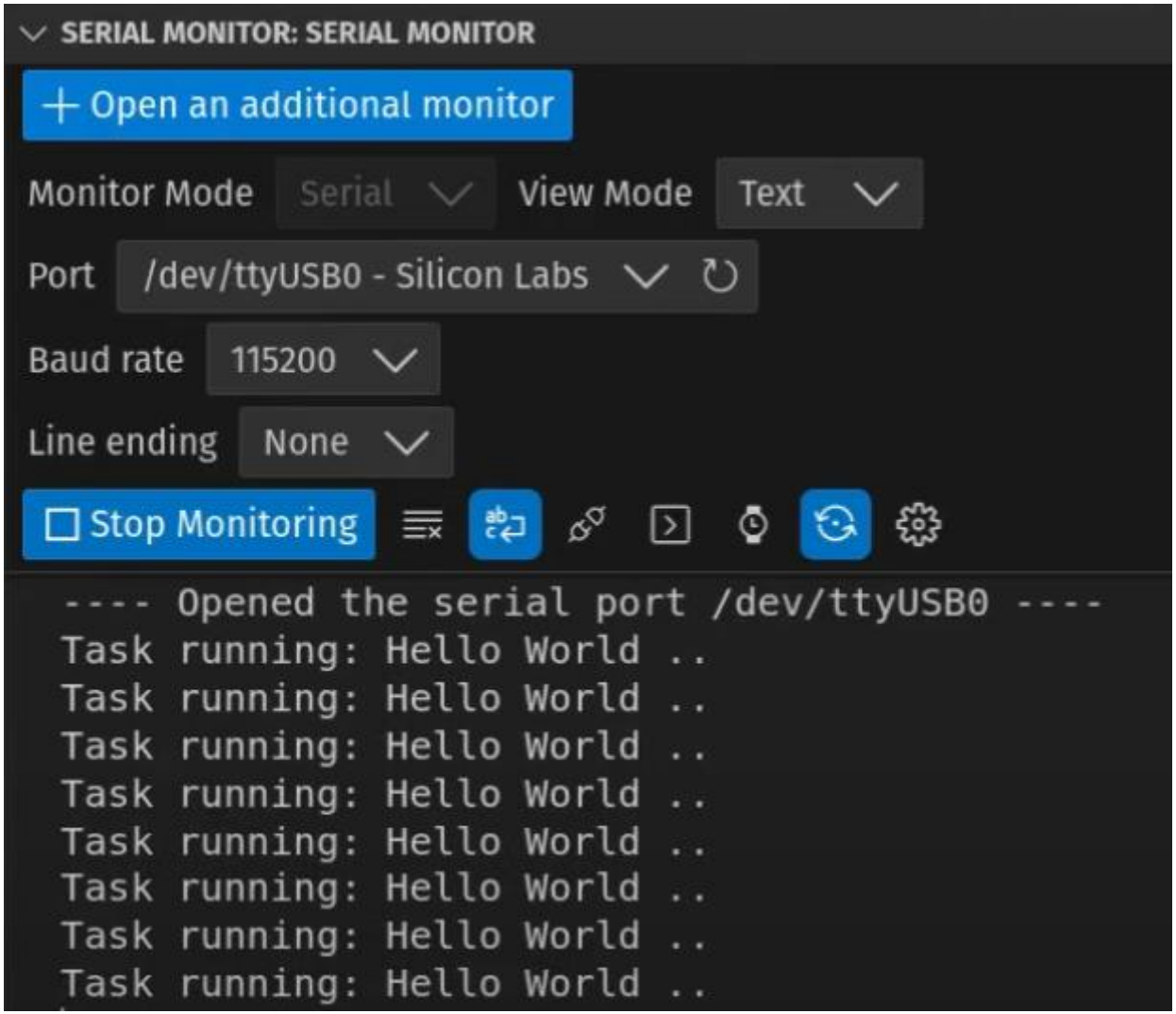
- ✓ Project Name: **example203**
- ✓ Enter Project directory: **D:\java-projects**
- ✓ Choose ESP-IDF Board: **ESP32-C3 chip(via ESP USB Bridge)**
- ✓ Choose serial port: COMx
- ✓ Choose Template: **template-app**

Bước 2: Sửa file **hello_world_main.c** như sau:

hello_world_main.c

```
#include <stdio.h>
#include <inttypes.h>
#include "sdkconfig.h"
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "esp_chip_info.h"
#include "esp_flash.h"
TaskHandle_t HelloWorldTaskHandle = NULL;
void HelloWorld_Task(void *arg)
{
    while (1)
    {
        printf("Task running: Hello World ..\n");
        vTaskDelay(1000 / portTICK_PERIOD_MS);
    }
}
void app_main(void)
{
    xTaskCreate(HelloWorld_Task, "HelloWorld", 4096, NULL, 10, &HelloWorldTaskHandle);
    //xTaskCreatePinnedToCore(HelloWorld_Task, "HelloWorld", 4096, NULL, 10, &HelloWorldTaskHandle, 1);  // Run on Core 1
}
```

Bước 3: Build project và nạp chương trình vào vi điều khiển


SERIAL MONITOR: SERIAL MONITOR
+ Open an additional monitor
Monitor Mode
Serial
View Mode
Text
Port
/dev/ttyUSB0 - Silicon Labs
Baud rate
115200
Line ending
None
Stop Monitoring
---- Opened the serial port /dev/ttyUSB0 ----
Task running: Hello World ..
Task running: Hello World ..
Task running: Hello World ..
Task running: Hello World ..
Task running: Hello World ..
Task running: Hello World ..
Task running: Hello World ..
Task running: Hello World ..

Example 2.04

Mục tiêu: Tạo và quản lý ứng dụng sử dụng FreeRTOS chạy trên vi điều khiển **ESP32**:

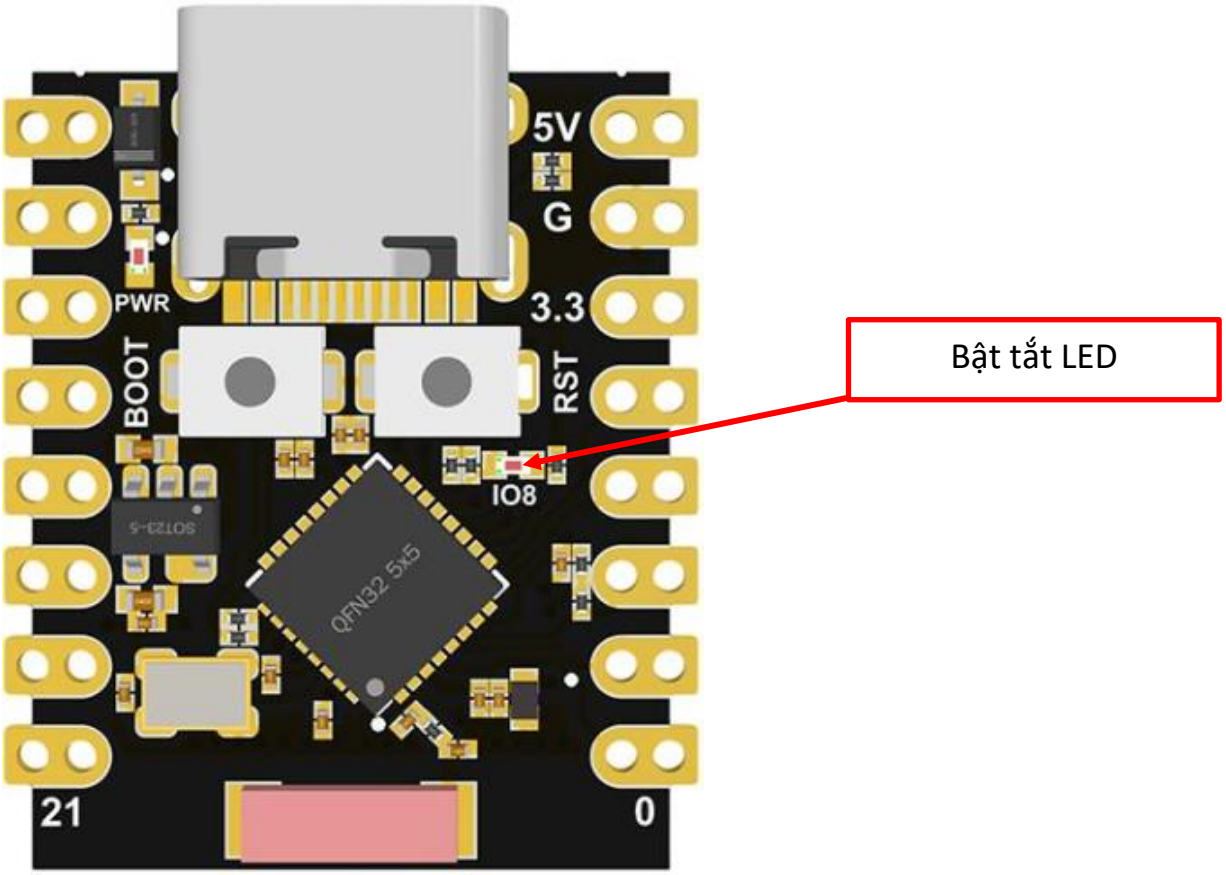
Yêu cầu: Ứng dụng thực hiện các chức năng sau:

- ✓ Thực hiện bật/tắt LED với chu kỳ 1000ms

Hướng dẫn:

Bước 1: Sử dụng Visual Studio Code, mở Command Palette (Ctrl+Shift+P) và nhập **IDF: New Project)** để bắt đầu tạo project với các thông tin như sau:

- ✓ Project Name: **example204**
- ✓ Enter Project directory: **D:\java-projects**
- ✓ Choose ESP-IDF Board: **ESP32-C3 chip(via ESP USB Bridge)**
- ✓ Choose serial port: COMx
- ✓ Choose Template: **template-app**



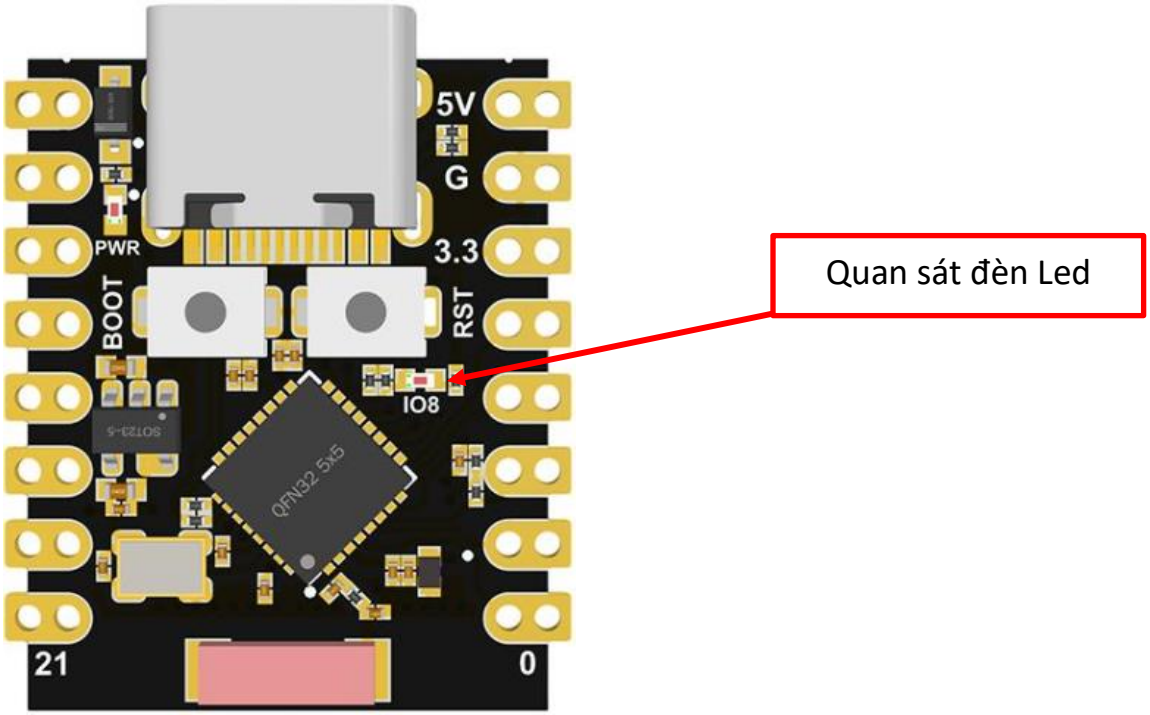
Bước 2: Sửa file **blink_main.c** như sau:

blink_main.c

```
#include <stdio.h>
#include <inttypes.h>
#include "sdkconfig.h"
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "esp_chip_info.h"
#include "esp_flash.h"

#include "driver/gpio.h"

#define BLINK_GPIO GPIO_NUM_8

TaskHandle_t BlinkyTaskHandle = NULL;

void Blinky_Task(void *arg)
{
    esp_rom_gpio_pad_select_gpio(BLINK_GPIO);
    gpio_set_direction(BLINK_GPIO, GPIO_MODE_OUTPUT);

    while (1)
    {
        gpio_set_level(BLINK_GPIO, 1);
        vTaskDelay(1000 / portTICK_PERIOD_MS);
        gpio_set_level(BLINK_GPIO, 0);
        vTaskDelay(1000 / portTICK_PERIOD_MS);
    }
}

void app_main(void)
{
```

```
    xTaskCreatePinnedToCore(Blinky_Task, "Blinky", 4096, NULL, 10, &BlinkyTaskHandle, 0); // Core 0
}
```

Bước 3: Build project và nạp chương trình vào vi điều khiển, sau đó quan sát trạng thái của bóng đèn LED trên board



Example 2.05

Mục tiêu: Tạo và quản lý GIPO. Thực hiện bật tắc đèn Led trên board ESP32-C3

Mục tiêu: Tạo và quản lý ứng dụng sử dụng FreeRTOS chạy trên vi điều khiển **ESP32**:

Yêu cầu: Ứng dụng thực hiện các chức năng sau:

- ✓ Thực hiện bật/tắt LED với chu kỳ 1000ms

Hướng dẫn:

Bước 1: Sử dụng Visual Studio Code, mở Command Palette (Ctrl+Shift+P) và nhập **IDF: New Project)** để bắt đầu tạo project với các thông tin như sau:

- ✓ Project Name: **example204**
- ✓ Enter Project directory: **D:\java-projects**
- ✓ Choose ESP-IDF Board: **ESP32-C3 chip(via ESP USB Bridge)**
- ✓ Choose serial port: COMx
- ✓ Choose Template: **template-app**

Hướng dẫn:

Bước 1: Sử dụng Visual Studio Code tạo Project với thông tin như sau:

- ✓ Project Name: **Lab02**
- ✓ Choose ESP-IDF Board: **ESP32-C3 chip (via ESP USB Bridge)**
- ✓ Enter Project directory: **D:\IoT-projects**

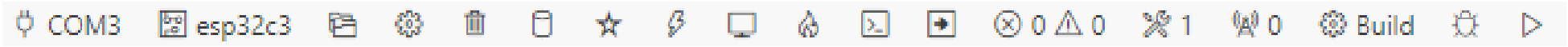
Bước 2: Sửa **main.c** file như sau:

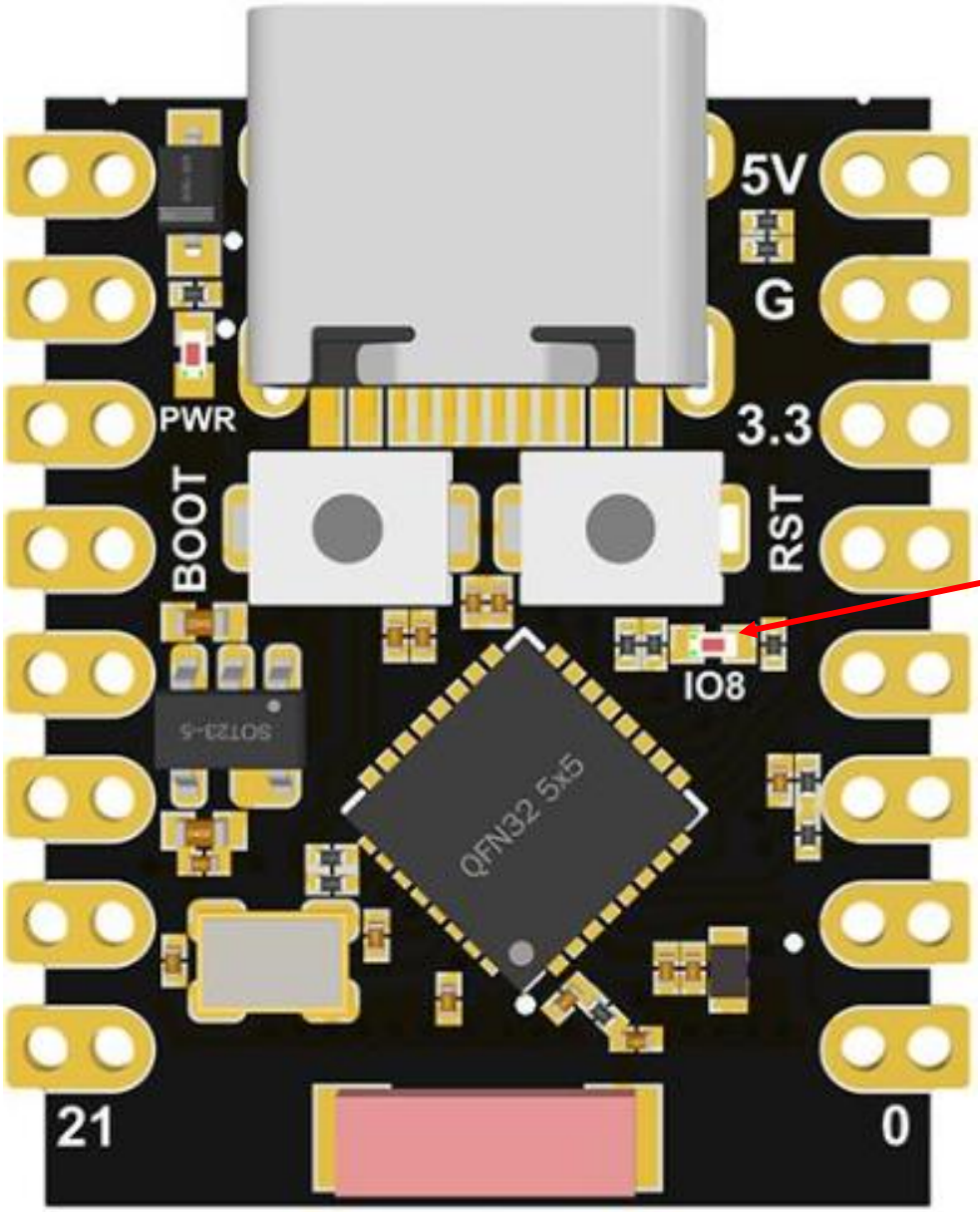
main.c

```
#include <stdio.h>

#include "driver\gpio.h"

#include "freeRTOS\freeRTOS.h"

#include "freeRTOS\task.h"


#define LED GPIO_NUM_8


void app_main(void)

{

    gpio_set_direction(LED, GPIO_MODE_DEF_OUTPUT);

    while(1)

    {

        gpio_set_level(LED, 1);

        vTaskDelay(100);

        gpio_set_level(LED, 0);

        vTaskDelay(100);

    }

}
```

Bước 3: Thực hiện build và flash chương trình lên board ESP32-C3

- ✓ Thực hiện mở Command Palette (Ctrl+Shift+P) sau đó chọn **ESP-IDF: Build your project** và **ESP-IDF: Flash (UART) your project**





Quan sát đèn Led

Example 2.06

Mục tiêu: Tạo và quản lý ứng dụng sử dụng FreeRTOS chạy trên vi điều khiển **ESP32**:

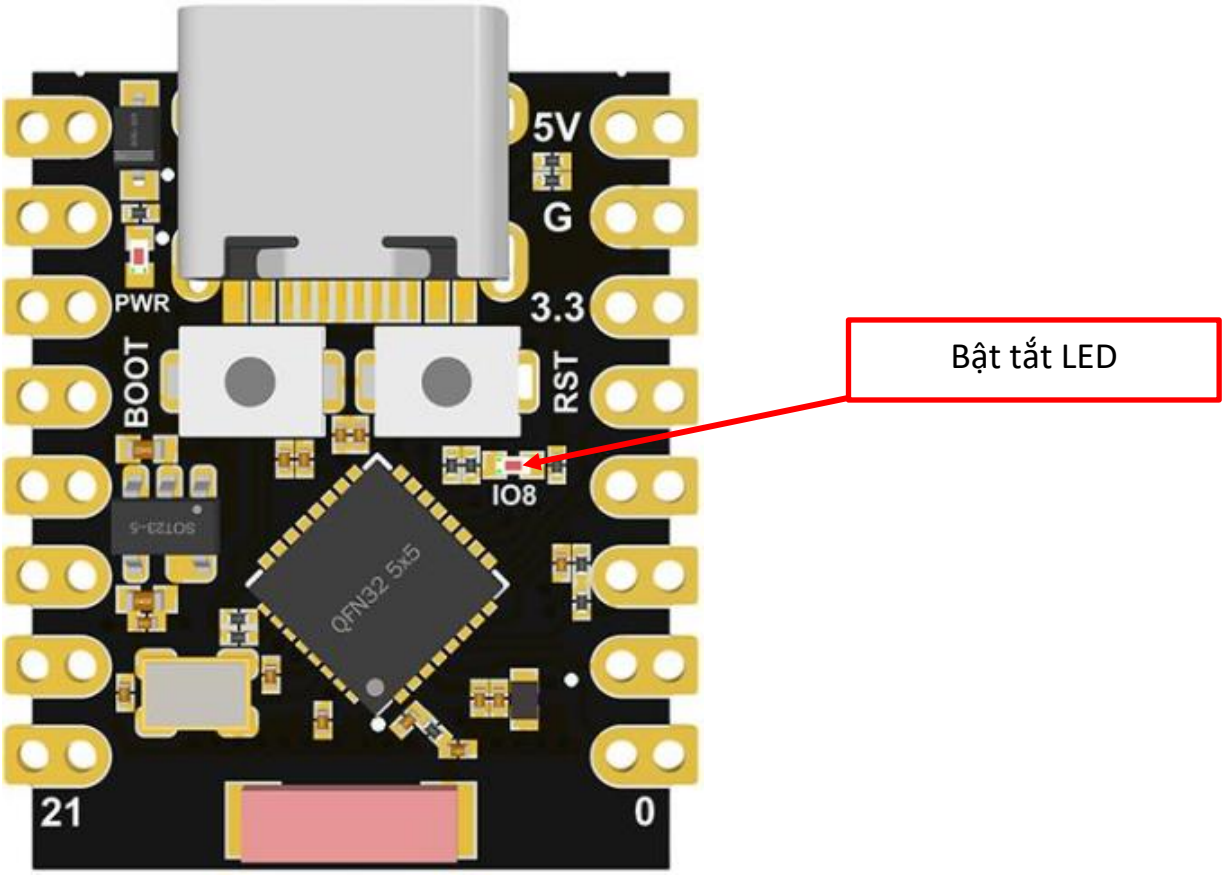
Yêu cầu: Ứng dụng thực hiện các chức năng sau:

- ✓ Thực hiện bật/tắt LED với chu kỳ 1000ms
- ✓ In ra Teminal nội dung Hello World ..

Hướng dẫn:

Bước 1: Sử dụng Visual Studio Code, mở Command Palette (Ctrl+Shift+P) và nhập **IDF: New Project)** để bắt đầu tạo project với các thông tin như sau:

- ✓ Project Name: **example205**
- ✓ Enter Project directory: **D:\java-projects**
- ✓ Choose ESP-IDF Board: **ESP32-C3 chip(via ESP USB Bridge)**
- ✓ Choose serial port: COMx



Bước 2: Sửa file **blink_main.c** như sau:

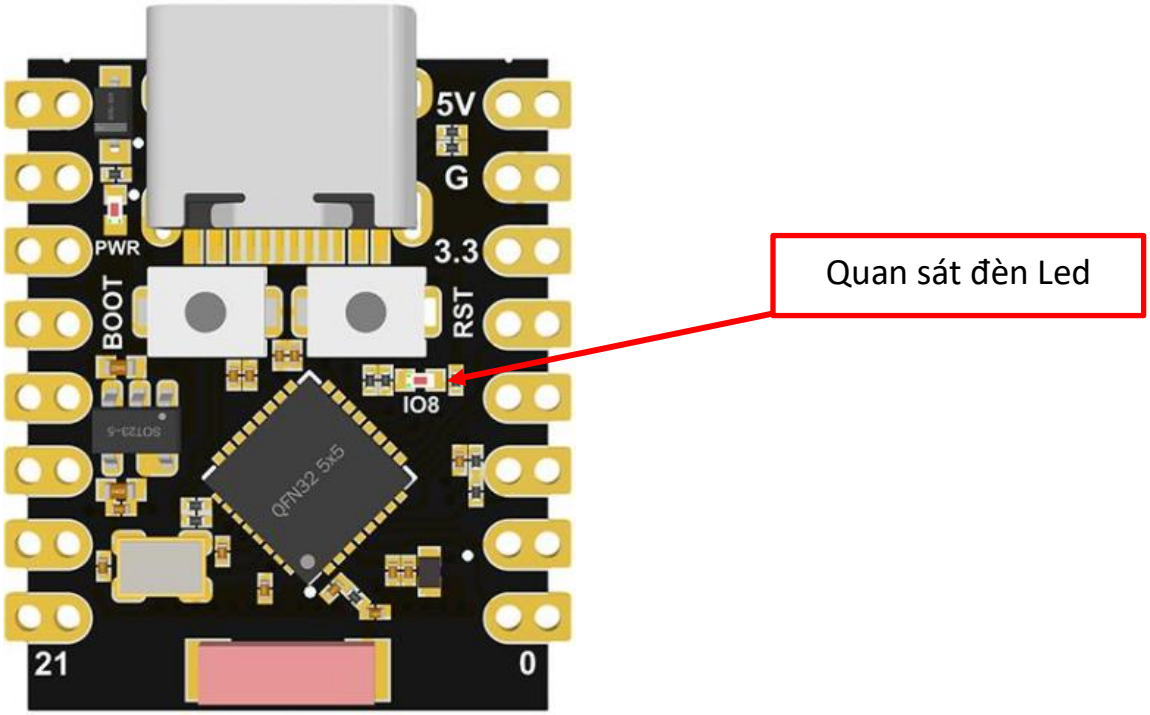
blink_main.c

```
#include <stdio.h>
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "driver/gpio.h"
#include "esp_log.h"
#include "sdkconfig.h"

#define BLINK_GPIO GPIO_NUM_8

TaskHandle_t HelloWorldTaskHandle = NULL;
TaskHandle_t BlinkyTaskHandle = NULL;

void HelloWorld_Task(void *arg)
{
    while (1)
    {
        printf("Task running: Hello World ..\n");
        vTaskDelay(1000 / portTICK_PERIOD_MS);
    }
}

void Blinky_Task(void *arg)
{
    esp_rom_gpio_pad_select_gpio(BLINK_GPIO);
    gpio_set_direction(BLINK_GPIO, GPIO_MODE_OUTPUT);

    int count_second = 0;

    while (1)
    {
```

```
        count_second += 1;

        switch (count_second)
        {
        case 10:
            vTaskSuspend(HelloWorldTaskHandle);
            printf("HelloWorld task suspended .. \n");
            break;
        case 14:
            vTaskResume(HelloWorldTaskHandle);
            printf("HelloWorld task resumed .. \n");
            break;
        case 20:
            vTaskDelete(HelloWorldTaskHandle);
            printf("HelloWorld task deleted .. \n");
            break;
        default:
            break;
        }
        gpio_set_level(BLINK_GPIO, 1);
        vTaskDelay(1000 / portTICK_PERIOD_MS);
        gpio_set_level(BLINK_GPIO, 0);
        vTaskDelay(1000 / portTICK_PERIOD_MS);
    }
}

void app_main(void)
{
    xTaskCreatePinnedToCore(Blinky_Task, "Blinky", 4096, NULL, 10, &BlinkyTaskHandle, 0);             // Core 0
    xTaskCreatePinnedToCore(HelloWorld_Task, "HelloWorld", 4096, NULL, 10, &HelloWorldTaskHandle, 1); // Core 1
}
```

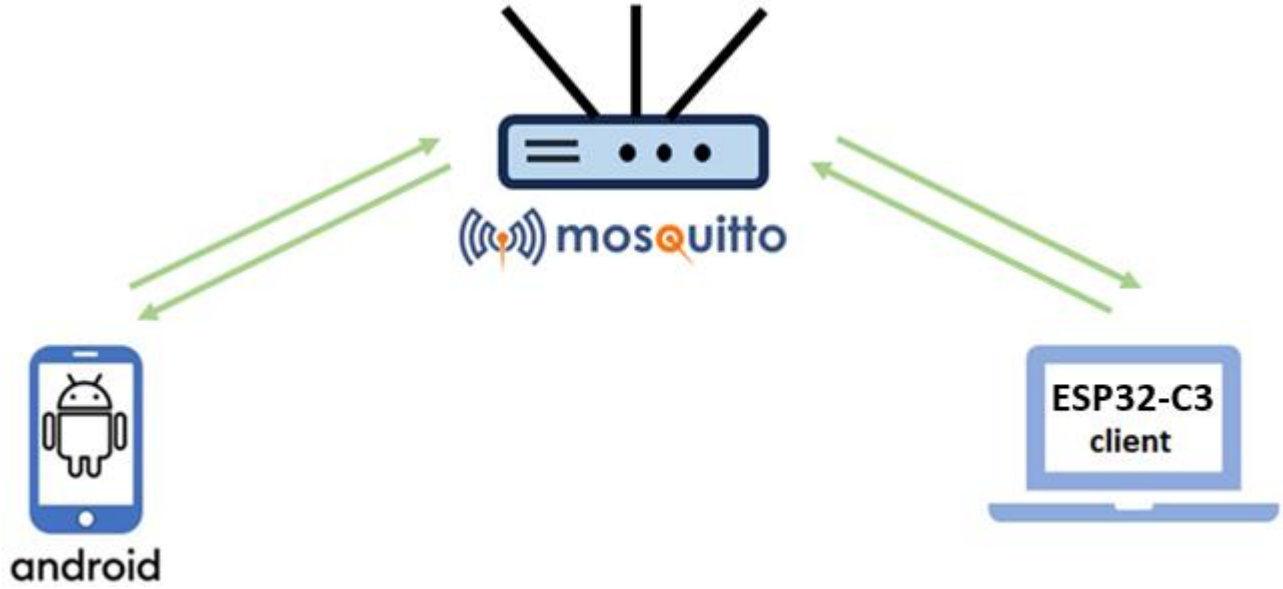
Bước 3: Build project và nạp chương trình vào vi điều khiển, sau đó quan sát trạng thái của bóng đèn LED trên board


ESP-IDF v5.1.2
COM1
esp32
0 0
Build


5V
G
3.3
PWR
BOOT
RST
IO8
QFN32 5x5
21
0
Quan sát đèn Led

Example 2.10

Mục tiêu: Tạo và quản lý ứng dụng sử dụng MQTT client chạy trên vi điều khiển **ESP32** (Sử dụng lệnh **idf.py menuconfig** trong ESP-IDF Terminal để cấu hình WIFI) có kiến trúc như sau:



Yêu cầu: Ứng dụng thực hiện các chức năng sau:

- ✓ Sử dụng **ESP32** Subscribe với **Broker** với topic có tên *"/topic/qos0*
- ✓ Sử dụng **ESP32** Publish nội dung "Hi from the IoT application" đến **Broker** với topic có tên *"/topic/qos1*

Hướng dẫn:

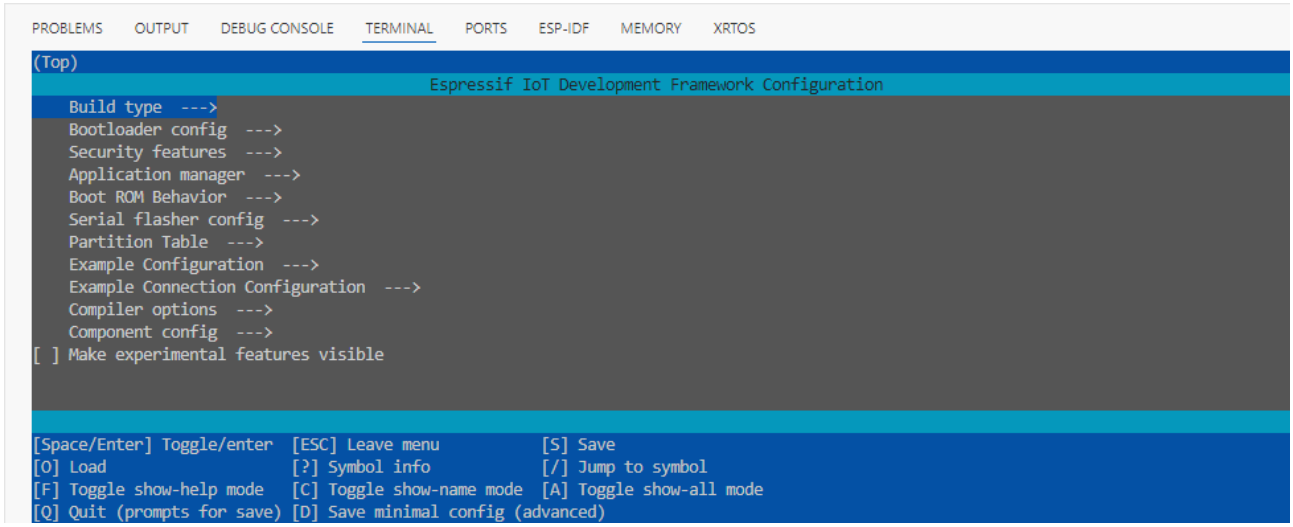
Bước 1: Cấu hình **MQTT Broker**

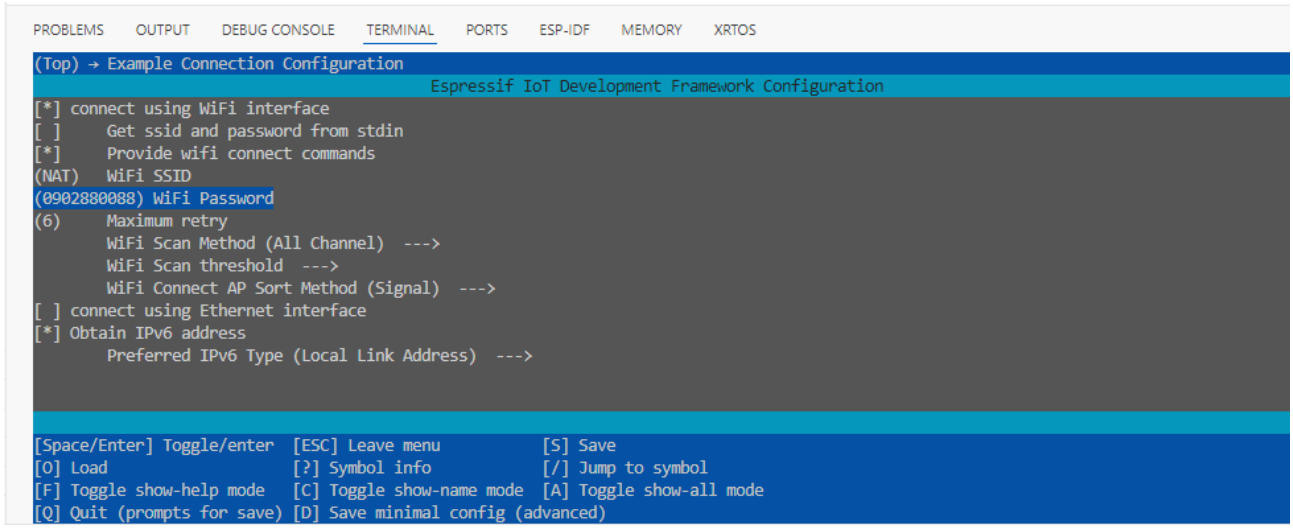
. . .\mosquitto\mosquitto.conf
`listener 1883` `allow_anonymous true`

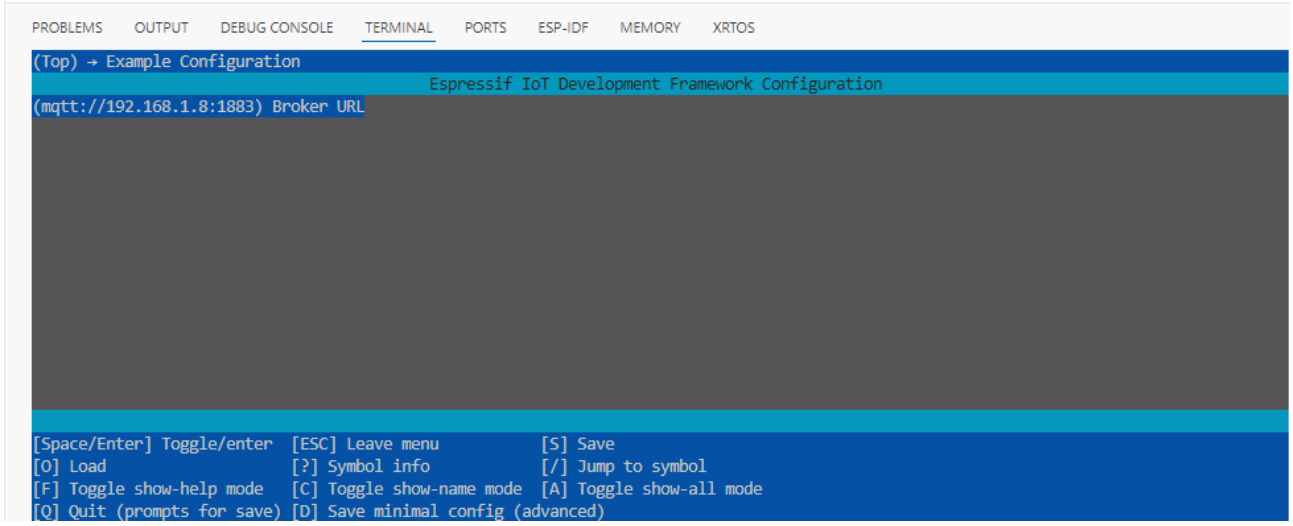
Bước 2: Sử dụng Visual Studio Code, mở Command Palette (Ctrl+Shift+P) và nhập **IDF: New Project)** để bắt đầu tạo project với các thông tin như sau:

- ✓ Project Name: **example210**
- ✓ Enter Project directory: **D:\java-projects**
- ✓ Choose ESP-IDF Board: **ESP32-C3 chip(via ESP USB Bridge)**
- ✓ Choose serial port: COMx

Bước 3: Sử dụng lệnh **idf.py menuconfig** trong ESP-IDF Terminal để cấu hình WIFI


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ESP-IDF MEMORY XRTOS
(Top)
Espressif IoT Development Framework Configuration
Build type --->
Bootloader config --->
Security features --->
Application manager --->
Boot ROM Behavior --->
Serial flasher config --->
Partition Table --->
Example Configuration --->
Example Connection Configuration --->
Compiler options --->
Component config --->
[ ] Make experimental features visible
[Space/Enter] Toggle/enter [ESC] Leave menu [S] Save
[O] Load [?] Symbol info [/] Jump to symbol
[F] Toggle show-help mode [C] Toggle show-name mode [A] Toggle show-all mode
[Q] Quit (prompts for save) [D] Save minimal config (advanced)


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ESP-IDF MEMORY XRTOS
(Top) → Example Connection Configuration
Espressif IoT Development Framework Configuration
[*] connect using WiFi interface
[ ] Get ssid and password from stdin
[*] Provide wifi connect commands
(NAT) WiFi SSID
(0902880088) WiFi Password
(6) Maximum retry
WiFi Scan Method (All Channel) --->
WiFi Scan threshold --->
WiFi Connect AP Sort Method (Signal) --->
[ ] connect using Ethernet interface
[*] Obtain IPv6 address
Preferred IPv6 Type (Local Link Address) --->
[Space/Enter] Toggle/enter [ESC] Leave menu [S] Save
[O] Load [?] Symbol info [/] Jump to symbol
[F] Toggle show-help mode [C] Toggle show-name mode [A] Toggle show-all mode
[Q] Quit (prompts for save) [D] Save minimal config (advanced)


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ESP-IDF MEMORY XRTOS
(Top) → Example Configuration
Espressif IoT Development Framework Configuration
(mqtt://192.168.1.8:1883) Broker URL
[Space/Enter] Toggle/enter [ESC] Leave menu [S] Save
[O] Load [?] Symbol info [/] Jump to symbol
[F] Toggle show-help mode [C] Toggle show-name mode [A] Toggle show-all mode
[Q] Quit (prompts for save) [D] Save minimal config (advanced)

Bước 4: Sửa file **app_main.c** như sau:

app_main.c

```
#include <stdio.h>
#include <stdint.h>
#include <stddef.h>
#include <string.h>
#include "esp_wifi.h"
#include "esp_system.h"
#include "nvs_flash.h"
#include "esp_event.h"
#include "esp_netif.h"
#include "protocol_examples_common.h"

#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "freertos/semphr.h"
#include "freertos/queue.h"

#include "lwip/sockets.h"
#include "lwip/dns.h"
#include "lwip/netdb.h"

#include "esp_log.h"
#include "mqtt_client.h"

static const char *TAG = "MQTT_EXAMPLE";

static void mqtt_event_handler(void *handler_args, esp_event_base_t base, int32_t event_id, void *event_data)
```

```
{
    ESP_LOGD(TAG, "Event dispatched from event loop base=%s, event_id=%" PRIi32 "", base, event_id);
    esp_mqtt_event_handle_t event = event_data;
    esp_mqtt_client_handle_t client = event->client;
    int msg_id;
    switch ((esp_mqtt_event_id_t)event_id)
    {
    case MQTT_EVENT_CONNECTED:
        ESP_LOGI(TAG, "MQTT_EVENT_CONNECTED");
        msg_id = esp_mqtt_client_publish(client, "/topic/qos1", "data_3", 0, 1, 0);
        ESP_LOGI(TAG, "sent publish successful, msg_id=%d", msg_id);

        msg_id = esp_mqtt_client_subscribe(client, "/topic/qos0", 0);
        ESP_LOGI(TAG, "sent subscribe successful, msg_id=%d", msg_id);

        msg_id = esp_mqtt_client_subscribe(client, "/topic/qos1", 1);
        ESP_LOGI(TAG, "sent subscribe successful, msg_id=%d", msg_id);

        msg_id = esp_mqtt_client_unsubscribe(client, "/topic/qos1");
        ESP_LOGI(TAG, "sent unsubscribe successful, msg_id=%d", msg_id);
        break;
    case MQTT_EVENT_DISCONNECTED:
        ESP_LOGI(TAG, "MQTT_EVENT_DISCONNECTED");
        break;

    case MQTT_EVENT_SUBSCRIBED:
        ESP_LOGI(TAG, "MQTT_EVENT_SUBSCRIBED, msg_id=%d", event->msg_id);
        msg_id = esp_mqtt_client_publish(client, "/topic/qos0", "data", 0, 0, 0);
        ESP_LOGI(TAG, "sent publish successful, msg_id=%d", msg_id);
```

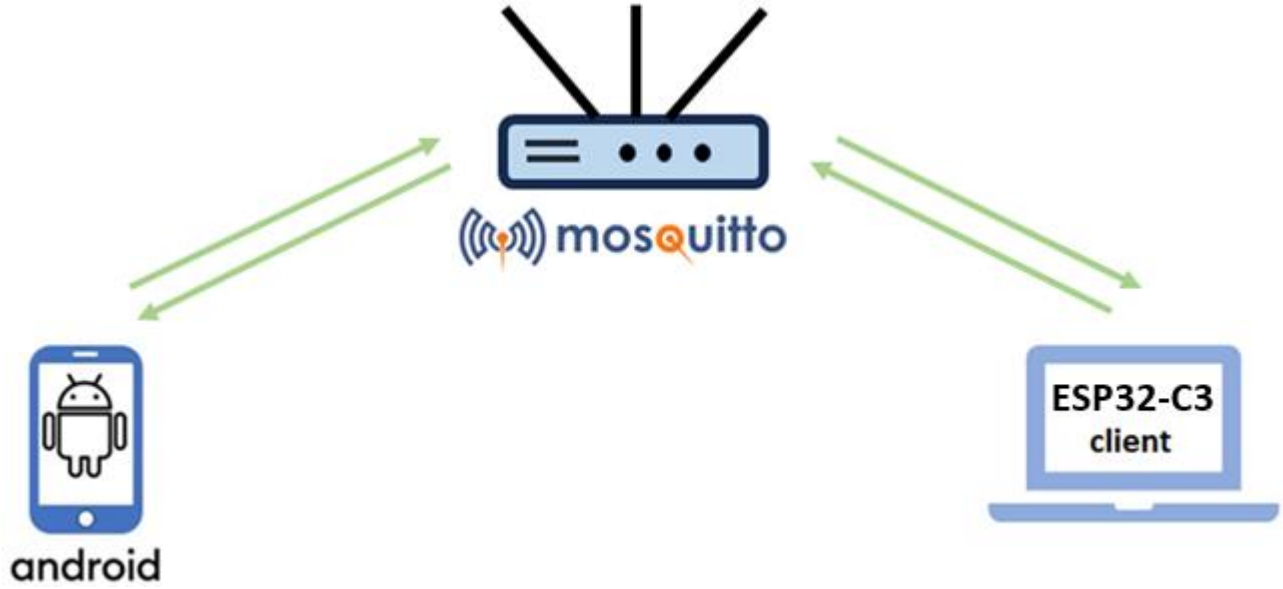
```
        break;
    case MQTT_EVENT_UNSUBSCRIBED:
        ESP_LOGI(TAG, "MQTT_EVENT_UNSUBSCRIBED, msg_id=%d", event->msg_id);
        break;
    case MQTT_EVENT_PUBLISHED:
        ESP_LOGI(TAG, "MQTT_EVENT_PUBLISHED, msg_id=%d", event->msg_id);
        break;
    case MQTT_EVENT_DATA:
        ESP_LOGI(TAG, "MQTT_EVENT_DATA");
        printf("TOPIC=%.*s\r\n", event->topic_len, event->topic);
        printf("DATA=%.*s\r\n", event->data_len, event->data);
        break;
    case MQTT_EVENT_ERROR:
        ESP_LOGI(TAG, "MQTT_EVENT_ERROR");
        break;
    default:
        ESP_LOGI(TAG, "Other event id:%d", event->event_id);
        break;
    }
}

static void mqtt_app_start(void)
{
    esp_mqtt_client_config_t mqtt_cfg = {
        .broker.address.uri = CONFIG_BROKER_URL,
    };

    esp_mqtt_client_handle_t client = esp_mqtt_client_init(&mqtt_cfg);
    esp_mqtt_client_register_event(client, ESP_EVENT_ANY_ID, mqtt_event_handler, NULL);
```

```
    esp_mqtt_client_start(client);
}

void app_main(void)
{
    ESP_LOGI(TAG, "[APP] Startup..");
    ESP_LOGI(TAG, "[APP] Free memory: %" PRIu32 " bytes", esp_get_free_heap_size());
    ESP_LOGI(TAG, "[APP] IDF version: %s", esp_get_idf_version());

    esp_log_level_set("*", ESP_LOG_INFO);
    esp_log_level_set("mqtt_client", ESP_LOG_VERBOSE);
    esp_log_level_set("MQTT_EXAMPLE", ESP_LOG_VERBOSE);
    esp_log_level_set("TRANSPORT_BASE", ESP_LOG_VERBOSE);
    esp_log_level_set("esp-tls", ESP_LOG_VERBOSE);
    esp_log_level_set("TRANSPORT", ESP_LOG_VERBOSE);
    esp_log_level_set("outbox", ESP_LOG_VERBOSE);

    ESP_ERROR_CHECK(nvs_flash_init());
    ESP_ERROR_CHECK(esp_netif_init());
    ESP_ERROR_CHECK(esp_event_loop_create_default());

    ESP_ERROR_CHECK(example_connect());

    mqtt_app_start();
}
```

Bước 5: Build project và nạp chương trình vào vi điều khiển

Example 2.11

Mục tiêu: Tạo và quản lý ứng dụng sử dụng MQTT client chạy trên vi điều khiển **ESP32** (không sử dụng **idf.py menuconfig** cấu hình WIFI code) có kiến trúc như sau:



Yêu cầu: Ứng dụng thực hiện các chức năng sau:

- ✓ Sử dụng **MQTT-Explorer** Subscribe với **Broker** với topic có tên */test/topic*
- ✓ Sử dụng **ESP32** Publish nội dung "`Hello AIoT`" đến **Broker** với topic có tên */test/topic*
- ✓ Sử dụng **MQTT-Explorer** publish tới **ESP32** nội dung **"Hi from the IoT application"** với topic có tên **/test/topic1**

Hướng dẫn:

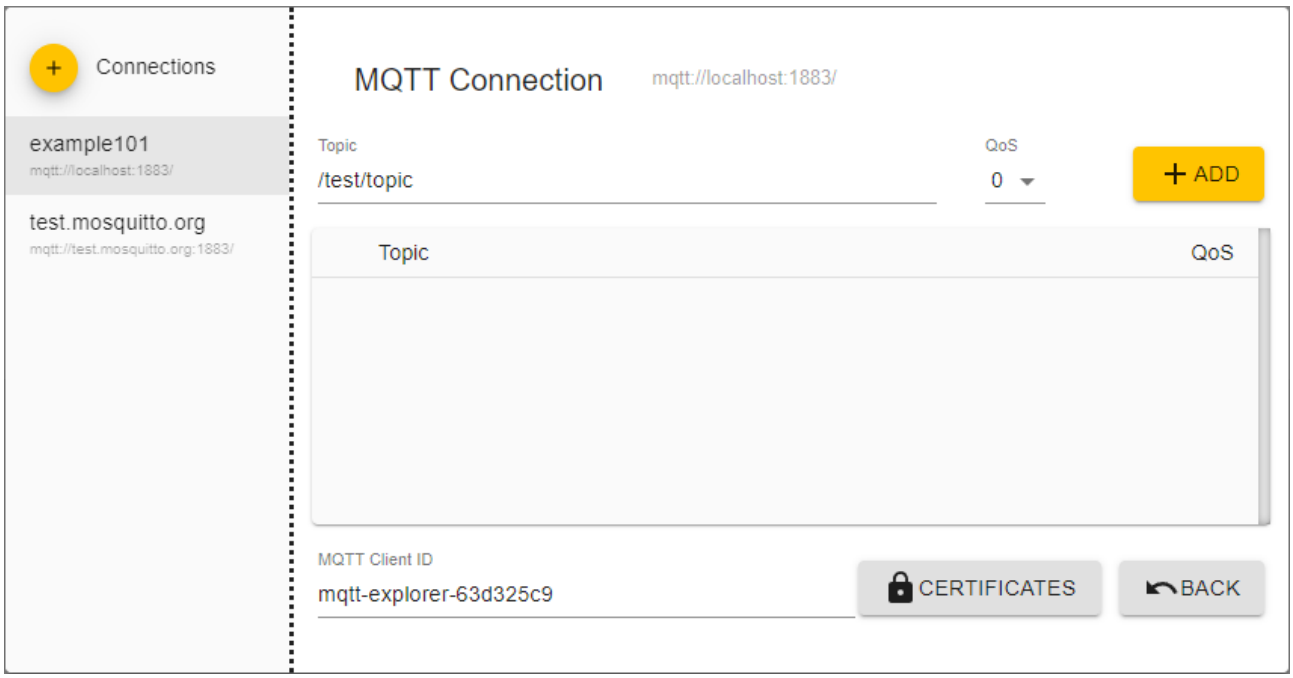
Bước 1: Cấu hình **MQTT Broker**

. . .\mosquitto\mosquitto.conf
`listener 1883` `allow_anonymous true`

Bước 2: Sử dụng Visual Studio Code, mở Command Palette (Ctrl+Shift+P) và nhập **IDF: New Project)** để bắt đầu tạo project với các thông tin như sau:

- ✓ Project Name: **example211**
- ✓ Enter Project directory: **D:\java-projects**
- ✓ Choose ESP-IDF Board: **ESP32-C3 chip(via ESP USB Bridge)**
- ✓ Choose serial port: COMx

Bước 3: Sử dụng **MQTT-Explorer** đăng ký topic có tên **/test/topic** với Broker


Connections
example101
mqtt://localhost:1883/
test.mosquitto.org
mqtt://test.mosquitto.org:1883/
MQTT Connection
mqtt://localhost:1883/
Topic
/test/topic
QoS
0
ADD
Topic
QoS
MQTT Client ID
mqtt-explorer-63d325c9
CERTIFICATES
BACK

Bước 4: Sửa file **app_main.c** như sau:

app_main.c

```
#include <stdio.h>
#include <stdint.h>
#include <stddef.h>
#include <string.h>
#include "esp_wifi.h"
#include "esp_system.h"
#include "nvs_flash.h"
#include "esp_event.h"
#include "esp_netif.h"
#include "protocol_examples_common.h"

#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "freertos/semphr.h"
#include "freertos/queue.h"

#include "lwip/sockets.h"
#include "lwip/dns.h"
#include "lwip/netdb.h"

#include "esp_log.h"
#include "mqtt_client.h"

static const char *TAG = "MQTT_EXAMPLE";
#define ESP_WIFI_SSID "YOUR_SSID"
#define ESP_WIFI_PASS "YOUR_PASSWORD"
#define ESP_BROKER_IP "BROKER_IP" //mqtt://192.168.1.4:1883
```

```
uint32_t MQTT_CONNECTED = 0;
static void mqtt_app_start(void);

static void wifi_event_handler(void *arg, esp_event_base_t event_base, int32_t event_id, void *event_data)
{
    if (event_base == WIFI_EVENT)
    {
        switch (event_id)
        {
        case WIFI_EVENT_STA_START:
            esp_wifi_connect();
            ESP_LOGI(TAG, "Trying to connect with Wi-Fi");
            break;
        case WIFI_EVENT_STA_DISCONNECTED:
            ESP_LOGI(TAG, "Disconnected: Retrying Wi-Fi");
            esp_wifi_connect();
            break;
        default:
            break;
        }
    }
    else if (event_base == IP_EVENT && event_id == IP_EVENT_STA_GOT_IP)
    {
        ESP_LOGI(TAG, "Got IP: Starting MQTT Client");
        mqtt_app_start();
    }
}
```

```
void wifi_init(void)
{
    ESP_ERROR_CHECK(esp_netif_init());
    ESP_ERROR_CHECK(esp_event_loop_create_default());
    esp_netif_create_default_wifi_sta();

    wifi_init_config_t cfg = WIFI_INIT_CONFIG_DEFAULT();
    ESP_ERROR_CHECK(esp_wifi_init(&cfg));

    ESP_ERROR_CHECK(esp_event_handler_instance_register(WIFI_EVENT, ESP_EVENT_ANY_ID, &wifi_event_handler, NULL, NULL));
    ESP_ERROR_CHECK(esp_event_handler_instance_register(IP_EVENT, IP_EVENT_STA_GOT_IP, &wifi_event_handler, NULL, NULL));

    wifi_config_t wifi_config = {
        .sta = {
            .ssid = ESP_WIFI_SSID,
            .password = ESP_WIFI_PASS,
            .threshold.authmode = WIFI_AUTH_WPA2_PSK,
        },
    };
    ESP_ERROR_CHECK(esp_wifi_set_mode(WIFI_MODE_STA));
    ESP_ERROR_CHECK(esp_wifi_set_config(WIFI_IF_STA, &wifi_config));
    ESP_ERROR_CHECK(esp_wifi_start());
}

static void mqtt_event_handler(void *handler_args, esp_event_base_t base, int32_t event_id, void *event_data)
{
    ESP_LOGD(TAG, "Event dispatched from event loop base=%s, event_id=%ld", base, (long)event_id);
    esp_mqtt_event_handle_t event = event_data;
    esp_mqtt_client_handle_t client = event->client;
```

```
    int msg_id;
    switch ((esp_mqtt_event_id_t)event_id)
    {
    case MQTT_EVENT_CONNECTED:
        ESP_LOGI(TAG, "MQTT_EVENT_CONNECTED");
        MQTT_CONNECTED = 1;
        msg_id = esp_mqtt_client_subscribe(client, "/test/topic1", 0);
        ESP_LOGI(TAG, "Subscribed, msg_id=%d", msg_id);
        break;
    case MQTT_EVENT_DISCONNECTED:
        ESP_LOGI(TAG, "MQTT_EVENT_DISCONNECTED");
        MQTT_CONNECTED = 0;
        break;
    case MQTT_EVENT_DATA:
        ESP_LOGI(TAG, "MQTT_EVENT_DATA");
        printf("TOPIC=%.*s\r\n", event->topic_len, event->topic);
        printf("DATA=%.*s\r\n", event->data_len, event->data);
        break;
    default:
        ESP_LOGI(TAG, "Other event id:%ld", (long)event->event_id);
        break;
    }
}

esp_mqtt_client_handle_t client = NULL;
static void mqtt_app_start(void)
{
    ESP_LOGI(TAG, "STARTING MQTT");
    esp_mqtt_client_config_t mqttConfig = {
```

```
        .broker.address.uri = ESP_BROKER_IP,
    };
    client = esp_mqtt_client_init(&mqttConfig);
    esp_mqtt_client_register_event(client, ESP_EVENT_ANY_ID, mqtt_event_handler, client);
    esp_mqtt_client_start(client);
}

void Publisher_Task(void *params)
{
    while (true)
    {
        if (MQTT_CONNECTED)
        {
            esp_mqtt_client_publish(client, "/test/topic", "Hello AIoT", 0, 0, 0);
            vTaskDelay(1500 / portTICK_PERIOD_MS);
        }
    }
}

void app_main(void)
{
    esp_err_t ret = nvs_flash_init();
    if (ret == ESP_ERR_NVS_NO_FREE_PAGES || ret == ESP_ERR_NVS_NEW_VERSION_FOUND)
    {
        ESP_ERROR_CHECK(nvs_flash_erase());
        ret = nvs_flash_init();
    }
    ESP_ERROR_CHECK(ret);
    wifi_init();
```

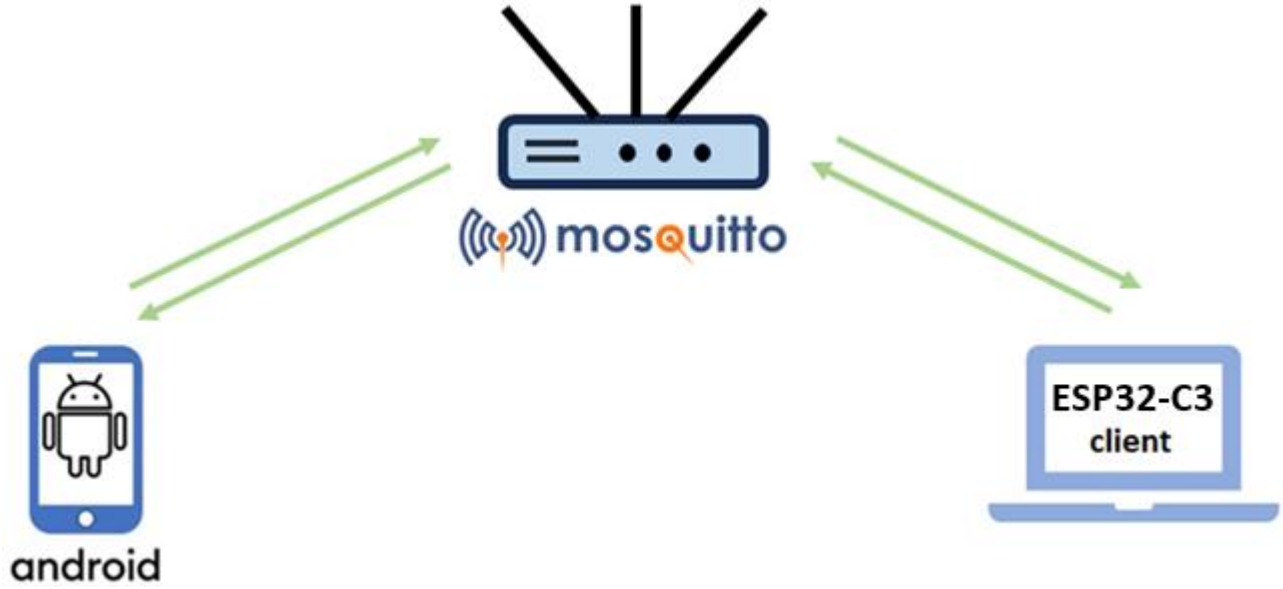
```
    xTaskCreate(Publisher_Task, "Publisher_Task", 1024 * 5, NULL, 5, NULL);
}
```

Bước 5: Build project và nạp chương trình vào vi điều khiển

Bước 6: Sử dụng **MQTT-Explorer** publish tới Broker nội dung **"Hi from the IoT application"** với topic có tên **/test/topic1**

Example 2.12

Mục tiêu: Tạo và quản lý ứng dụng sử dụng MQTT client chạy trên vi điều khiển **ESP32 (**cấu hình WIFI qua App thiết bi di động**)** có kiến trúc như sau:



Yêu cầu: Ứng dụng thực hiện các chức năng sau:

- ✓ Sử dụng **MQTT-Explorer** Subscribe với **Broker** với topic có tên */test/topic*

Hướng dẫn:

Bước 1: Cấu hình **MQTT Broker**

. . .\mosquitto\mosquitto.conf
`listener 1883` `allow_anonymous true`

Bước 2: Sử dụng Visual Studio Code, mở Command Palette (Ctrl+Shift+P) và nhập **IDF: New Project)** để bắt đầu tạo project với các thông tin như sau:

- ✓ Project Name: **example212**
- ✓ Enter Project directory: **D:\java-projects**
- ✓ Choose ESP-IDF Board: **ESP32-C3 chip(via ESP USB Bridge)**
- ✓ Choose serial port: COMx

Bước 3: Build app **EspTouch** từ source code:

- ✓ Android: https://github.com/EspressifApp/EsptouchForAndroid
- ✓ iOS: https://github.com/EspressifApp/EsptouchForIOS

Bước 4: Sửa file **app_main.c** như sau:

app_main.c

```
#include <string.h>
#include <stdlib.h>
#include "freertos/FreeRTOS.h"
#include "freertos/task.h"
#include "freertos/event_groups.h"
#include "esp_wifi.h"
#include "esp_eap_client.h"
#include "esp_event.h"
#include "esp_log.h"
#include "esp_system.h"
#include "nvs_flash.h"
#include "esp_netif.h"
#include "esp_smartconfig.h"
#include "esp_mac.h"

static EventGroupHandle_t s_wifi_event_group;

static const int CONNECTED_BIT = BIT0;
static const int ESPTOUCH_DONE_BIT = BIT1;
static const char *TAG = "smartconfig_example";

static void smartconfig_example_task(void *parm);

static void event_handler(void *arg, esp_event_base_t event_base,
                          int32_t event_id, void *event_data)
{
    if (event_base == WIFI_EVENT && event_id == WIFI_EVENT_STA_START)
```

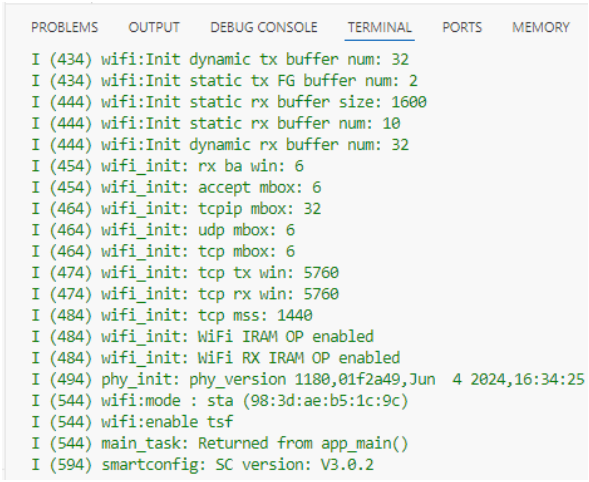
```
    {
        xTaskCreate(smartconfig_example_task, "smartconfig_example_task", 4096, NULL, 3, NULL);
    }
    else if (event_base == WIFI_EVENT && event_id == WIFI_EVENT_STA_DISCONNECTED)
    {
        esp_wifi_connect();
        xEventGroupClearBits(s_wifi_event_group, CONNECTED_BIT);
    }
    else if (event_base == IP_EVENT && event_id == IP_EVENT_STA_GOT_IP)
    {
        xEventGroupSetBits(s_wifi_event_group, CONNECTED_BIT);
    }
    else if (event_base == SC_EVENT && event_id == SC_EVENT_SCAN_DONE)
    {
        ESP_LOGI(TAG, "Scan done");
    }
    else if (event_base == SC_EVENT && event_id == SC_EVENT_FOUND_CHANNEL)
    {
        ESP_LOGI(TAG, "Found channel");
    }
    else if (event_base == SC_EVENT && event_id == SC_EVENT_GOT_SSID_PSWD)
    {
        ESP_LOGI(TAG, "Got SSID and password");

        smartconfig_event_got_ssid_pswd_t *evt = (smartconfig_event_got_ssid_pswd_t *)event_data;
        wifi_config_t wifi_config;
        uint8_t ssid[33] = {0};
        uint8_t password[65] = {0};
        uint8_t rvd_data[33] = {0};
```

```
        bzero(&wifi_config, sizeof(wifi_config_t));
        memcpy(wifi_config.sta.ssid, evt->ssid, sizeof(wifi_config.sta.ssid));
        memcpy(wifi_config.sta.password, evt->password, sizeof(wifi_config.sta.password));

#ifdef CONFIG_SET_MAC_ADDRESS_OF_TARGET_AP
        wifi_config.sta.bssid_set = evt->bssid_set;
        if (wifi_config.sta.bssid_set == true)
        {
            ESP_LOGI(TAG, "Set MAC address of target AP: " MACSTR " ", MAC2STR(evt->bssid));
            memcpy(wifi_config.sta.bssid, evt->bssid, sizeof(wifi_config.sta.bssid));
        }
#endif

        memcpy(ssid, evt->ssid, sizeof(evt->ssid));
        memcpy(password, evt->password, sizeof(evt->password));
        ESP_LOGI(TAG, "SSID:%s", ssid);
        ESP_LOGI(TAG, "PASSWORD:%s", password);
        if (evt->type == SC_TYPE_ESPTOUCH_V2)
        {
            ESP_ERROR_CHECK(esp_smartconfig_get_rvd_data(rvd_data, sizeof(rvd_data)));
            ESP_LOGI(TAG, "RVD_DATA:");
            for (int i = 0; i < 33; i++)
            {
                printf("%02x ", rvd_data[i]);
            }
            printf("\n");
        }
```

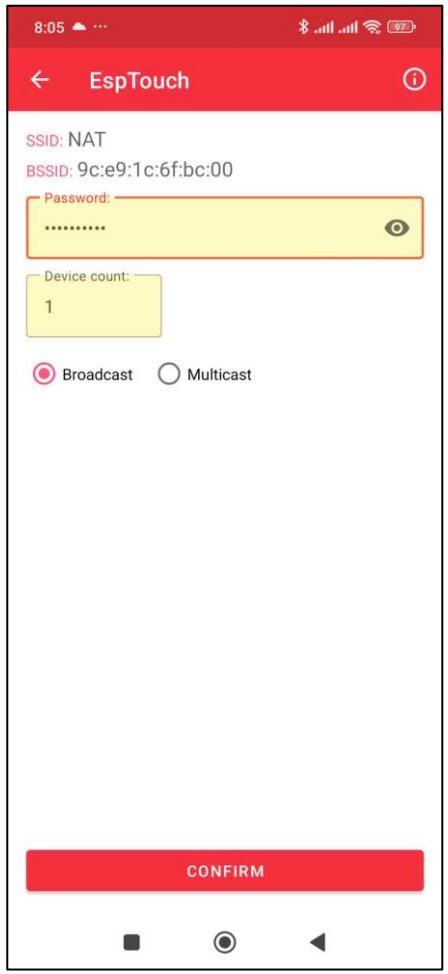
```
        ESP_ERROR_CHECK(esp_wifi_disconnect());
        ESP_ERROR_CHECK(esp_wifi_set_config(WIFI_IF_STA, &wifi_config));
        esp_wifi_connect();
    }
    else if (event_base == SC_EVENT && event_id == SC_EVENT_SEND_ACK_DONE)
    {
        xEventGroupSetBits(s_wifi_event_group, ESPTOUCH_DONE_BIT);
    }
}

static void initialise_wifi(void)
{
    ESP_ERROR_CHECK(esp_netif_init());
    s_wifi_event_group = xEventGroupCreate();
    ESP_ERROR_CHECK(esp_event_loop_create_default());
    esp_netif_t *sta_netif = esp_netif_create_default_wifi_sta();
    assert(sta_netif);

    wifi_init_config_t cfg = WIFI_INIT_CONFIG_DEFAULT();
    ESP_ERROR_CHECK(esp_wifi_init(&cfg));

    ESP_ERROR_CHECK(esp_event_handler_register(WIFI_EVENT, ESP_EVENT_ANY_ID, &event_handler, NULL));
    ESP_ERROR_CHECK(esp_event_handler_register(IP_EVENT, IP_EVENT_STA_GOT_IP, &event_handler, NULL));
    ESP_ERROR_CHECK(esp_event_handler_register(SC_EVENT, ESP_EVENT_ANY_ID, &event_handler, NULL));

    ESP_ERROR_CHECK(esp_wifi_set_mode(WIFI_MODE_STA));
    ESP_ERROR_CHECK(esp_wifi_start());
}
```

```
static void smartconfig_example_task(void *parm)
{
    EventBits_t uxBits;
    ESP_ERROR_CHECK(esp_smartconfig_set_type(SC_TYPE_ESPTOUCH));
    smartconfig_start_config_t cfg = SMARTCONFIG_START_CONFIG_DEFAULT();
    ESP_ERROR_CHECK(esp_smartconfig_start(&cfg));
    while (1)
    {
        uxBits = xEventGroupWaitBits(s_wifi_event_group, CONNECTED_BIT | ESPTOUCH_DONE_BIT, true, false, portMAX_DELAY);
        if (uxBits & CONNECTED_BIT)
        {
            ESP_LOGI(TAG, "WiFi Connected to ap");
        }
        if (uxBits & ESPTOUCH_DONE_BIT)
        {
            ESP_LOGI(TAG, "smartconfig over");
            esp_smartconfig_stop();
            vTaskDelete(NULL);
        }
    }
}

void app_main(void)
{
    ESP_ERROR_CHECK(nvs_flash_init());
    initialise_wifi();
}
```

Bước 5: Build project và nạp chương trình vào vi điều khiển, sau đó theo dõi tại màn hình Terminal chúng ta sẽ thấy như sau:



Bước 6: Sử dụng App **EspTouch** trên Android nhập **Password** của WIFI → **CONFIRM** → sau đó theo dõi tại màn hình Terminal ESP32 sẽ kết nối WiFi với **SSID** và **Password** được gửi từ app




```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  MEMORY  XRTOS  ESP-IDF
I (5414) smartconfig: Start to find channel...
I (5414) smartconfig_example: Scan done
I (725404) smartconfig: TYPE: ESPTOUCH
I (725404) smartconfig: T|PHONE MAC:b2:88:67:9b:d2:cf
I (725414) smartconfig: T|AP MAC:9c:e9:1c:6f:bc:00
I (725414) smartconfig: Found channel on 4-0. Start to get ssid and password...
I (725414) smartconfig_example: Found channel
I (733694) smartconfig: T|pswd: 0902880088
I (733694) smartconfig: T|ssid: NAT
I (733694) smartconfig: T|bssid: 9c:e9:1c:6f:bc:00
I (733694) wifi:ic_disable_sniffer
I (733704) smartconfig_example: Got SSID and password
I (733704) smartconfig_example: Set MAC address of target AP: 9c:e9:1c:6f:bc:00
I (733714) smartconfig_example: SSID:NAT
I (733724) smartconfig_example: PASSWORD:0902880088
W (733724) wifi:Password length matches WPA2 standards, authmode threshold changes from OPEN to WPA2
I (733744) wifi:new:<4,0>, old:<4,0>, ap:<255,255>, sta:<4,0>, prof:1, snd_ch_cfg:0x0
I (733744) wifi:state: init -> auth (0xb0)
I (733754) wifi:state: auth -> assoc (0x0)
I (733764) wifi:state: assoc -> run (0x10)
I (735864) wifi:connected with NAT, aid = 9, channel 4, BW20, bssid = 9c:e9:1c:6f:bc:00
I (735864) wifi:security: WPA2-PSK, phy: bgn, rssi: -65
I (735864) wifi:pm start, type: 1

I (735864) wifi:dp: 1, bi: 102400, li: 3, scale listen interval from 307200 us to 307200 us
I (735874) wifi:set rx beacon pti, rx_bcn_pti: 0, bcn_timeout: 25000, mt_pti: 0, mt_time: 10000
I (735894) wifi:AP's beacon interval = 102400 us, DTIM period = 1
I (737914) wifi:<ba-add>idx:0 (ifx:0, 9c:e9:1c:6f:bc:00), tid:0, ssn:0, winSize:64
I (738024) wifi:<ba-add>idx:1 (ifx:0, 9c:e9:1c:6f:bc:00), tid:6, ssn:4, winSize:64
I (738884) esp_netif_handlers: sta ip: 192.168.1.8, mask: 255.255.255.0, gw: 192.168.1.1
```